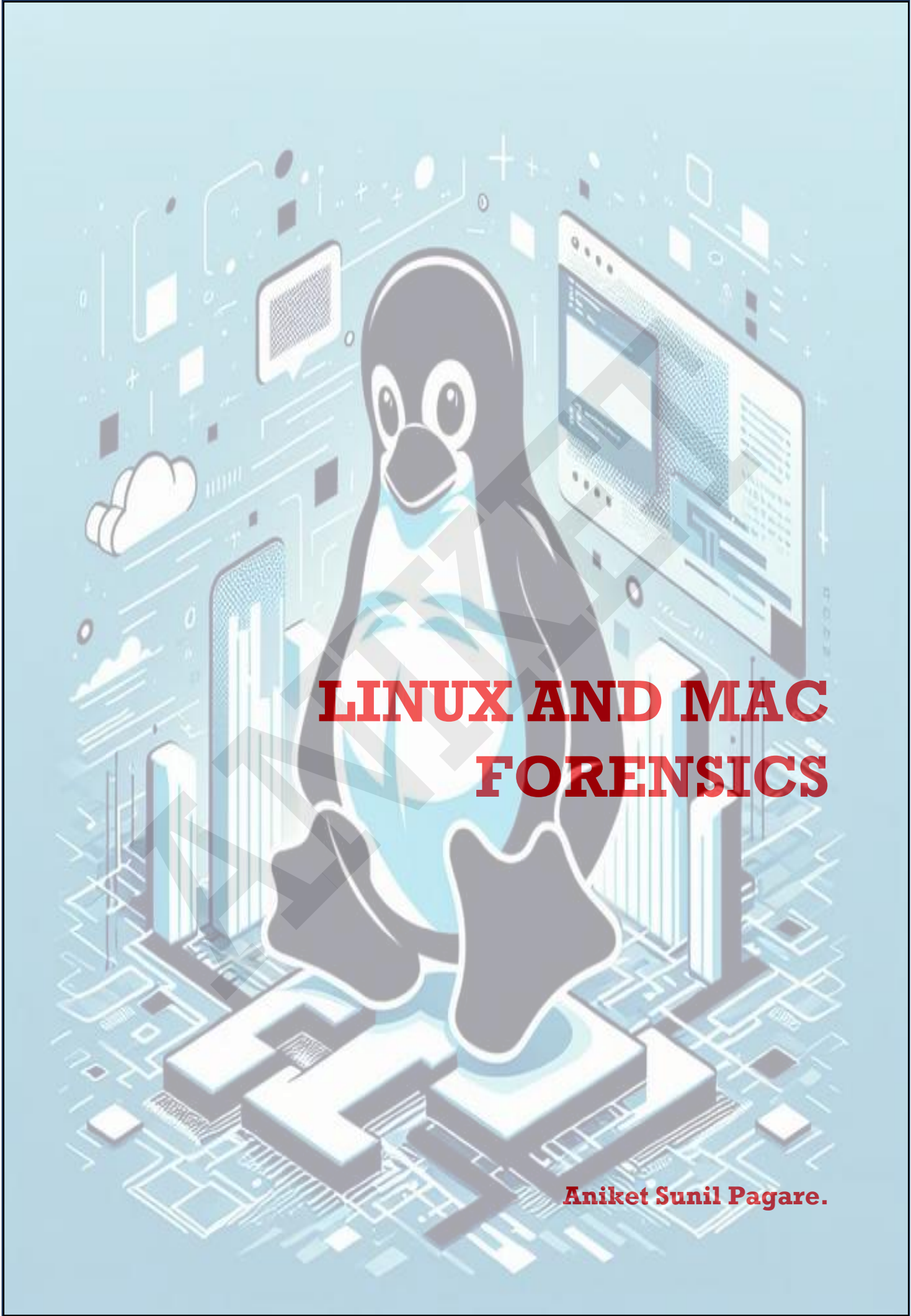




LINUX AND MAC FORENSICS

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Linux Forensics

Linux Forensics is the process of identifying, preserving, analyzing, and presenting digital evidence from a Linux-based system. It involves investigating logs, user activity, file changes, malware, rootkits, and other anomalies in the Linux environment.

Key Components of Linux Forensics

1. Disk Forensics

- Mount & analyze Linux partitions (ext4, xfs, btrfs)
 - Analyze deleted files and file metadata
 - Disk imaging: dd, dcfldd, guymager
-

2. Memory Forensics

- Capture memory using LiME
 - Analyze using Volatility or Rekall
 - Detect hidden processes, malware, kernel modules
-

3. Log Analysis

- /var/log/auth.log, /var/log/syslog, /var/log/messages
 - SSH logins, sudo activities, cron jobs
-

4. User Activity

- Bash history: ~/.bash_history
 - .ssh/authorized_keys, .ssh/known_hosts
 - Analyze cron jobs: crontab -l
-

5. File System Analysis

- Analyze inode timestamps (MAC times)
 - extundelete to recover deleted files
 - Hashing and integrity checks: md5sum, sha256sum
-

6. Network Forensics

- Capture packets: tcpdump, wireshark
 - Analyze open ports: netstat, lsof, ss
 - Check firewall logs: iptables, ufw logs
-

7. Rootkit and Malware Detection

- chkrootkit, rkhunter
 - Hidden processes or modules
 - YARA rules to detect malware signatures
-

8. Artifact Collection

- Installed packages: dpkg -l, rpm -qa
 - Kernel modules: lsmod
 - System configuration files: /etc/passwd, /etc/shadow, /etc/group
-

Common Tools in Linux Forensics

Tool	Use
dd	Bit-by-bit disk imaging
LiME	RAM acquisition on Linux
Volatility	RAM analysis (with Linux profile)

Tool	Use
Autopsy	GUI-based forensics platform
Sleuth Kit	File system analysis
Bulk Extractor	Extract email, credit cards, URLs, etc.
extundelete	Recover deleted files from ext3/ext4
chkrootkit	Rootkit scanner
rkhunter	Rootkit hunter
log2timeline	Create timelines from logs
plaso	Advanced timeline creation
binwalk	Analyze firmware images
foremost	File recovery
hashdeep	Recursive hash computation for directories
Wireshark	Packet capturing and analysis

Typical Linux Forensics Workflow

1. Preparation

- Create forensic image using dd or guymager
- Mount image read-only: `mount -o ro,loop disk.dd`
- `/mnt/forensic`

2. Initial Triage

- Identify system info, time, users
 - Check logs for suspicious activity
-

3. Deep Analysis

- Memory: vol.py -f mem.lime linux.pslist
 - File system metadata: fls, icat (from Sleuth Kit)
 - Timeline: log2timeline.py, plaso
-

4. Artifact Analysis

- Bash history, crontabs
 - Sudo logs, SSH keys
 - Detect malicious binaries
-

5. Reporting

- Maintain chain of custody
 - Write detailed report
 - Provide evidence for legal/investigative action
-

Important Artifacts to Analyze

- **/etc/shadow** – Hashed passwords
 - **~/.bash_history** – User commands
 - **/var/log/auth.log** – Login records
 - **/var/log/syslog** – System messages
 - **/etc/rc.local, /etc/init.d** – Persistent backdoors
 - **Hidden files** – ls -la, find / -name ".*"
-

Mac Forensics

Introduction to Mac Forensics

Mac forensics involves analyzing Apple macOS systems (MacBooks, iMacs, etc.) to uncover digital evidence. Since macOS is Unix-based, it has both similarities and differences compared to Windows or Linux forensics.

Key Areas of Mac Forensics

1. File System Forensics

- macOS primarily uses **APFS** (Apple File System) and older **HFS+**.
 - Forensic tools must understand APFS's container-based structure, snapshots, and metadata.
-

2. User Account and Authentication

- Analyze `/Users` directory
 - Passwords stored in **Keychain**
 - Examine **TCC.db** (Privacy database)
 - System Integrity Protection (SIP) impacts access
-

3. System Logs

- Located in `/var/log/`, `~/Library/Logs/`, and `/Library/Logs/`
 - Use `Console.app` or command-line tools (e.g., `log`, `syslog`)
-

4. Plist Files (Property Lists)

- macOS stores configuration and user activity in `.plist` files (in XML or binary)
- Common locations

- /Library/Preferences/
 - ~/Library/Preferences/
 - Analyze with plutil or Plist editors
-

5. Browser Artifacts

- Safari: ~/Library/Safari/
 - Chrome: ~/Library/Application Support/Google/Chrome/
 - Firefox: ~/Library/Application Support/Firefox/
-

6. Email Artifacts

- Apple Mail: ~/Library/Mail/
 - Outlook or Thunderbird may have unique formats
-

7. Chat & Messaging

- iMessage: chat.db in ~/Library/Messages/
 - Can extract messages, attachments, timestamps
-

8. Spotlight Indexing

- Analyze .Spotlight-V100 to find deleted or moved files
-

9. Time Machine Backups

- Analyze old snapshots
 - Located in
/Volumes/com.apple.TimeMachine.localsnapshots
-

10. Application Analysis

- macOS apps are stored in /Applications/

- Preferences often stored in .plist files

11. Removable Media & USB Artifacts

- system.log and /var/db/ log USB plug-ins
- diskutil list to view drives

12. Scheduled Jobs and Persistence

- **LaunchDaemons:** /Library/LaunchDaemons/
- **LaunchAgents:** ~/Library/LaunchAgents/
- Analyze for malware or persistence mechanisms

13. Deleted File Recovery

- APFS snapshot analysis or disk image recovery
- Carving tools like PhotoRec, Scalpel

Popular Mac Forensics Tools

Tool	Purpose
Magnet AXIOM	Comprehensive Mac analysis
BlackLight (by BlackBag)	Deep-dive into APFS, iMessages, Safari
MacQuisition	Live acquisition and imaging
Autopsy/Sleuth Kit	Open-source forensics
X-Ways Forensics	Can parse HFS+/APFS
ExifTool	Metadata extraction
iBackupBot	iPhone/iPad backups
sqlitebrowser	Analyze chat.db, other SQLite files

Tool	Purpose
Plist Editor Pro	For binary/XML plist analysis

📁 Common Artifacts and Locations

Artifact	Path
Messages	~/Library/Messages/chat.db
Safari History	~/Library/Safari/History.db
Chrome History	~/Library/Application Support/Google/Chrome/Default/History
Installed Apps	/Applications/
User Preferences	~/Library/Preferences/
Logs	/var/log/
Keychain (Credentials)	~/Library/Keychains/

📦 Imaging macOS

- Use **MacQuisition** or **FTK Imager** for creating a disk image
- Consider using .dmg, .E01, or raw image format

⚠️ Challenges in Mac Forensics

- **APFS encryption**
- **System Integrity Protection (SIP)** blocks access to some files
- **T2 Chip** and **Apple Silicon (M1/M2/M3)** restrict live booting or imaging
- Plist files in **binary** form
- Limited tools compared to Windows/Linux

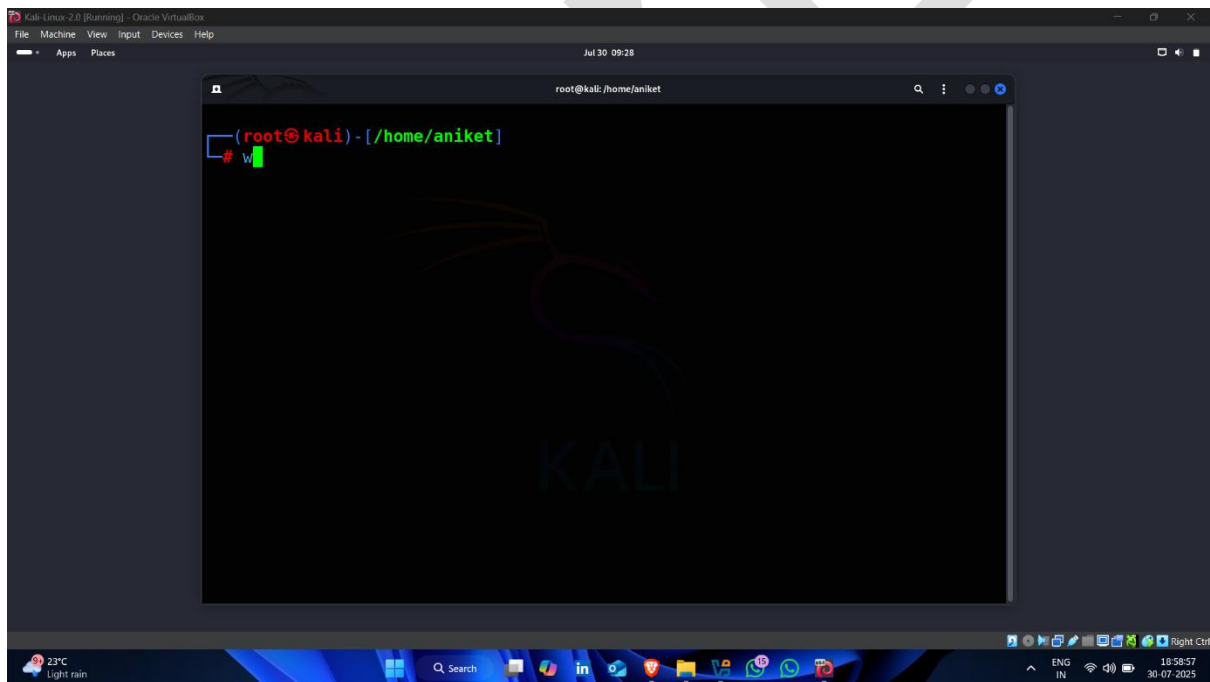
Lab-1 -:Acquire Volatile Data In linux System

Acquiring volatile data in a Linux system refers to the process of collecting temporary, dynamic data that resides in the system's memory or active processes. This kind of data is lost when the system is shut down or rebooted, so it's crucial to acquire it **before powering off or restarting the system during an investigation.**

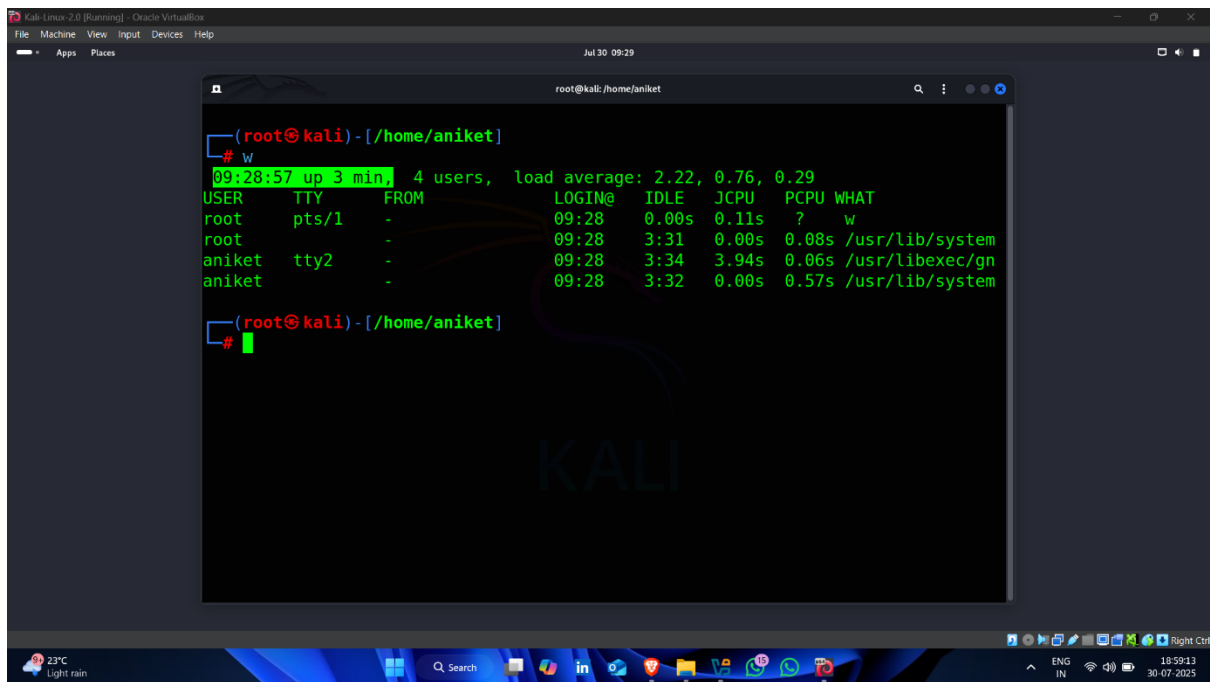
A) Acquire Volatile Data In linux System Using Kali Linux

Use some Following command for acquiring volatile data

Commands – W -- The w command shows who is logged in and what they are doing.



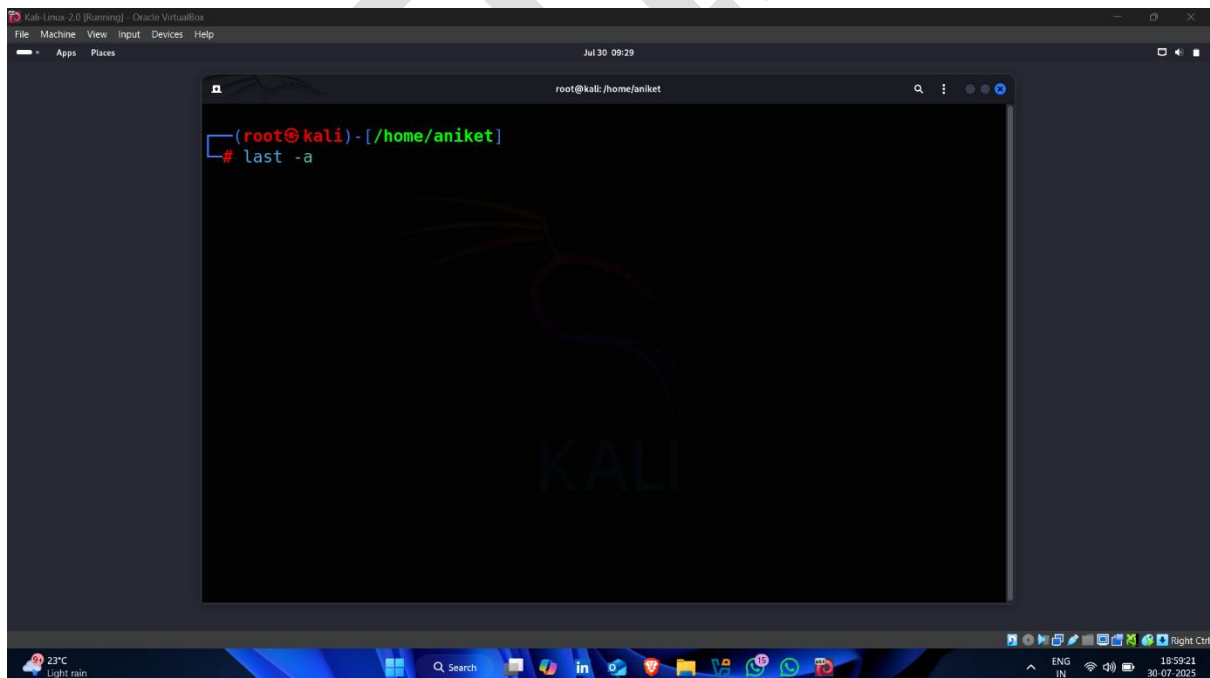
- Result 🖱️



The screenshot shows a terminal window in a Kali Linux virtual machine. The user is root at the kali machine, in the directory /home/aniket. They have entered the 'w' command, which displays system status and active sessions.

```
(root@kali) - [/home/aniket]
# w
09:28:57 up 3 min, 4 users, load average: 2.22, 0.76, 0.29
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU WHAT
root      pts/1    -                09:28   0.00s  0.11s  ?    w
root      -        -                09:28   3:31   0.00s  0.08s  /usr/lib/system
aniket    tty2     -                09:28   3:34   3.94s  0.06s  /usr/libexec/gn
aniket    -        -                09:28   3:32   0.00s  0.57s  /usr/lib/system
```

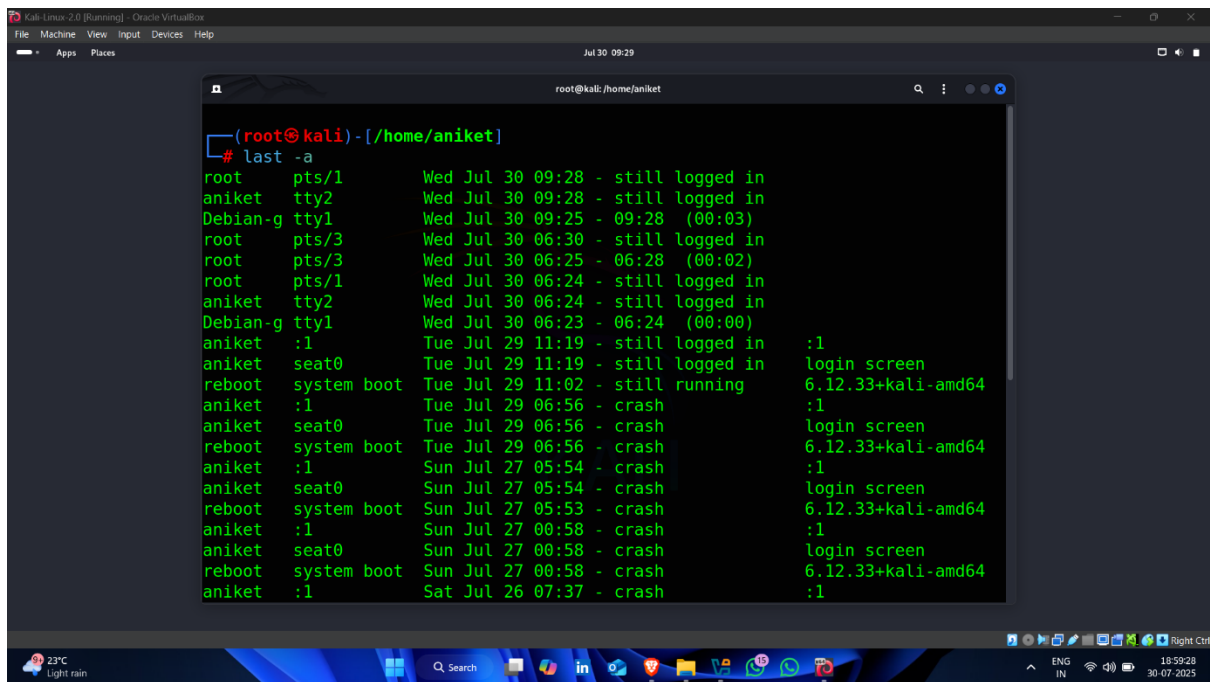
Command -: `last -a` -- is used to show the login history of users on a Linux system, including their IP addresses or hostnames.



The screenshot shows the same terminal window as before, but now the user has entered the command 'last -a'. The output of this command is not yet visible in the image.

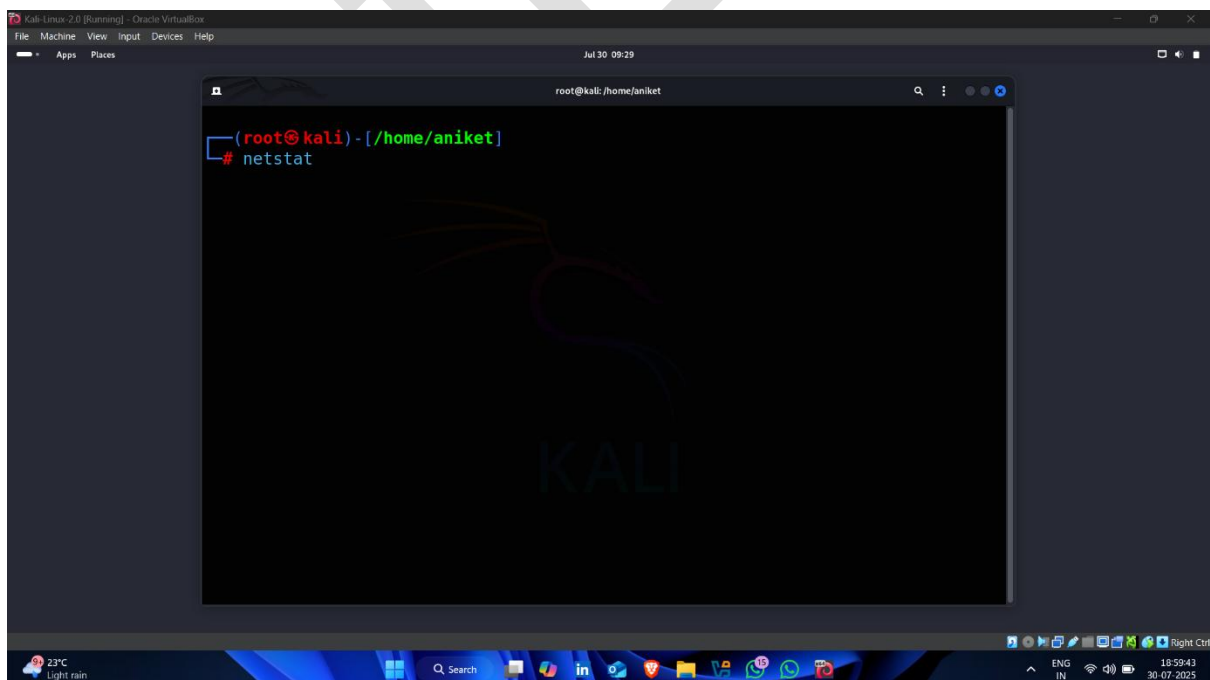
```
(root@kali) - [/home/aniket]
# last -a
```

- Result 🖐



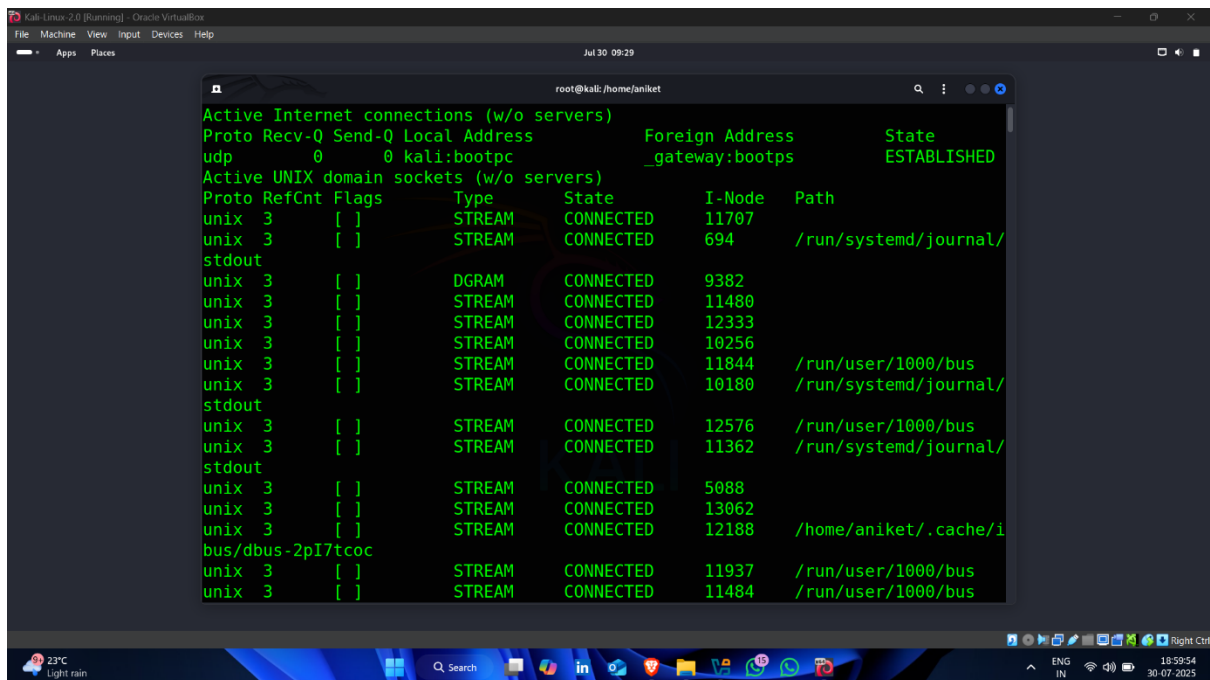
```
(root@kali) - [/home/aniket]
# last -a
root pts/1      Wed Jul 30 09:28 - still logged in
aniket tty2      Wed Jul 30 09:28 - still logged in
Debian-g tty1    Wed Jul 30 09:25 - 09:28 (00:03)
root pts/3      Wed Jul 30 06:30 - still logged in
root pts/3      Wed Jul 30 06:25 - 06:28 (00:02)
root pts/1      Wed Jul 30 06:24 - still logged in
aniket tty2      Wed Jul 30 06:24 - still logged in
Debian-g tty1    Wed Jul 30 06:23 - 06:24 (00:00)
aniket :1        Tue Jul 29 11:19 - still logged in      :1
aniket seat0    Tue Jul 29 11:19 - still logged in      login screen
reboot system boot Tue Jul 29 11:02 - still running        6.12.33+kali-amd64
aniket :1        Tue Jul 29 06:56 - crash                :1
aniket seat0    Tue Jul 29 06:56 - crash                login screen
reboot system boot Tue Jul 29 06:56 - crash        6.12.33+kali-amd64
aniket :1        Sun Jul 27 05:54 - crash                :1
aniket seat0    Sun Jul 27 05:54 - crash                login screen
reboot system boot Sun Jul 27 05:53 - crash        6.12.33+kali-amd64
aniket :1        Sun Jul 27 00:58 - crash                :1
aniket seat0    Sun Jul 27 00:58 - crash                login screen
reboot system boot Sun Jul 27 00:58 - crash        6.12.33+kali-amd64
aniket :1        Sat Jul 26 07:37 - crash                :1
```

Command – Netstat -- The netstat command is used to display network connections, routing tables, interface statistics, and listening ports in a Linux system.



```
(root@kali) - [/home/aniket]
# netstat
```

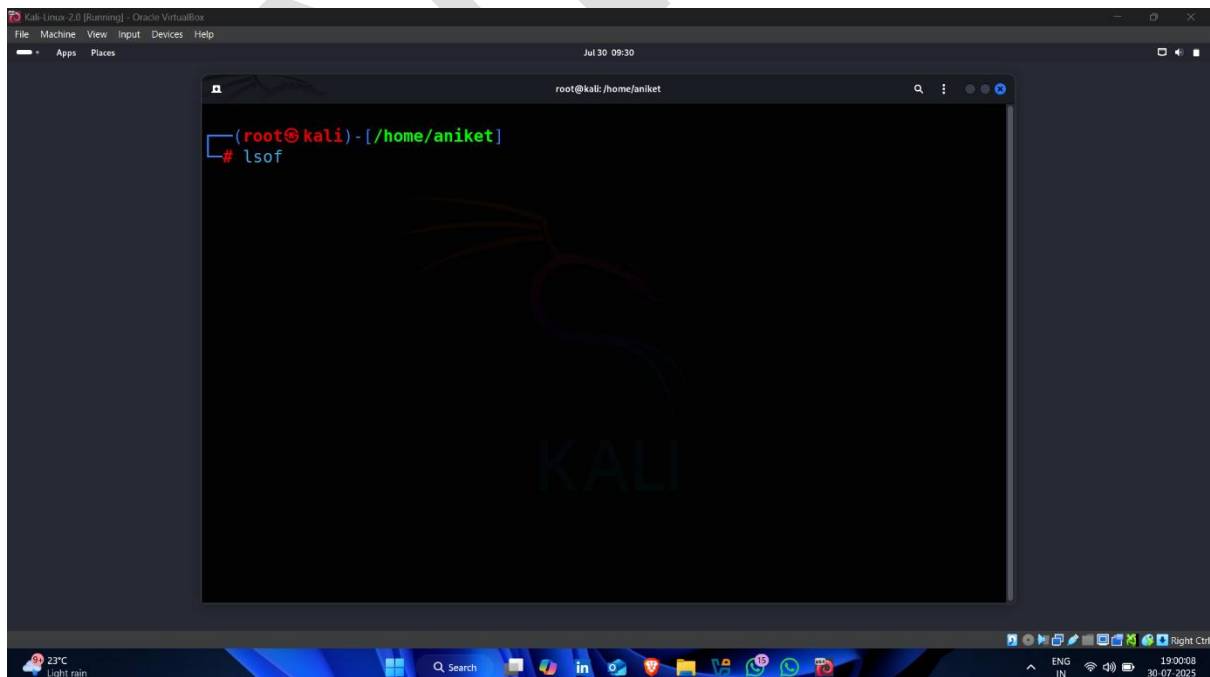
- Result 🖱️



```
root@kali: /home/aniket
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 kali:bootpc            _gateway:bootps        ESTABLISHED

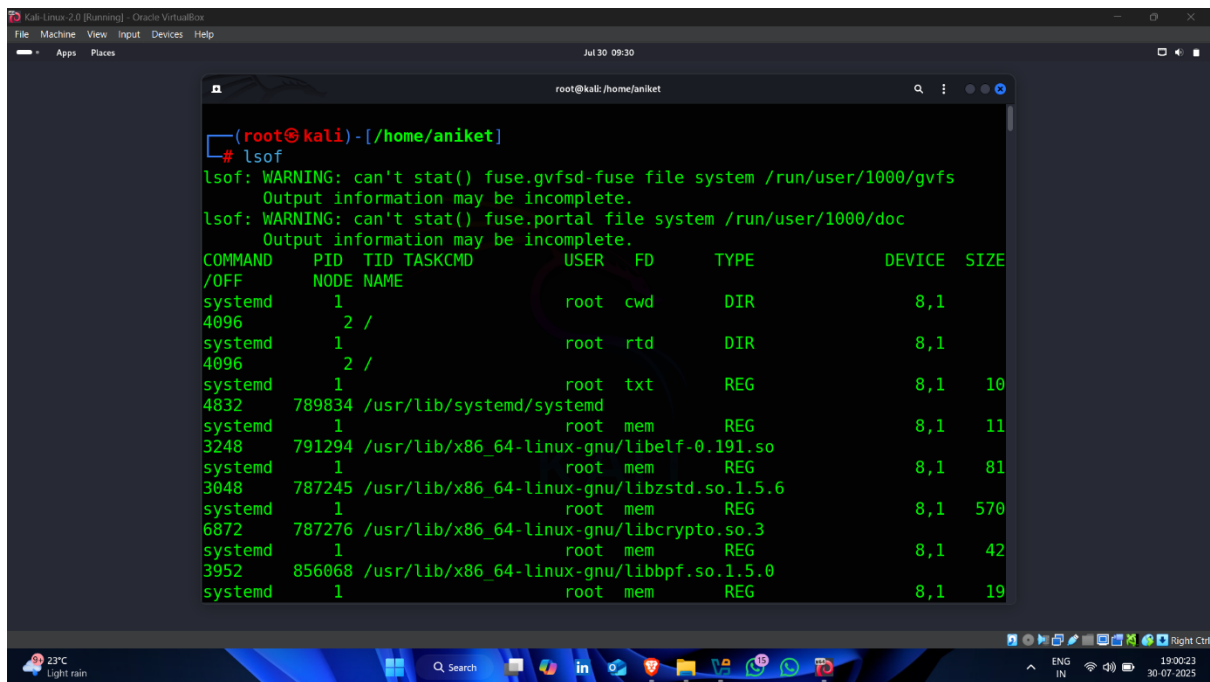
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State         I-Node  Path
unix  3      [ ]     STREAM    CONNECTED    11707
unix  3      [ ]     STREAM    CONNECTED    694      /run/systemd/journal/
stdout
unix  3      [ ]     DGRAM     CONNECTED    9382
unix  3      [ ]     STREAM    CONNECTED    11480
unix  3      [ ]     STREAM    CONNECTED    12333
unix  3      [ ]     STREAM    CONNECTED    10256
unix  3      [ ]     STREAM    CONNECTED    11844     /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    10180     /run/systemd/journal/
stdout
unix  3      [ ]     STREAM    CONNECTED    12576     /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    11362     /run/systemd/journal/
stdout
unix  3      [ ]     STREAM    CONNECTED    5088
unix  3      [ ]     STREAM    CONNECTED    13062
unix  3      [ ]     STREAM    CONNECTED    12188     /home/aniket/.cache/i
bus/dbus-2pI7tcoc
unix  3      [ ]     STREAM    CONNECTED    11937     /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    11484     /run/user/1000/bus
```

Command – lsof -- The lsof (List Open Files) command in Linux is used to list information about files opened by processes. Since in Linux everything is a file (including sockets, pipes, devices), lsof gives powerful insight into running processes and system activity.



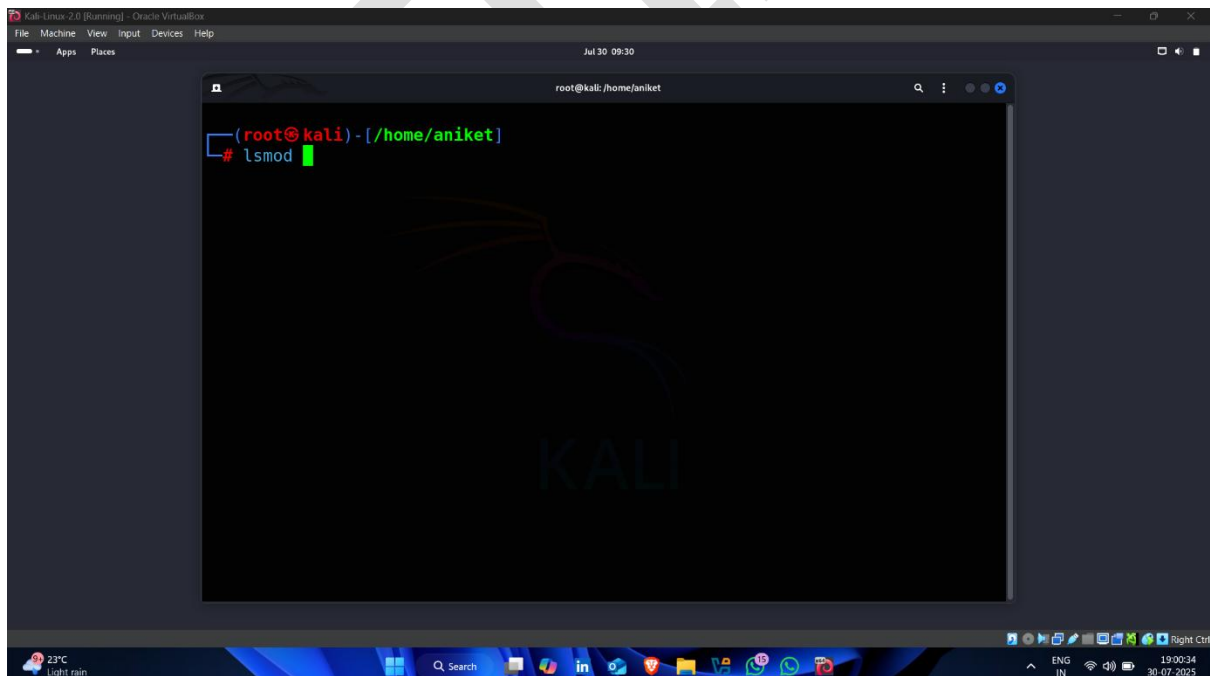
```
(root@kali) - [/home/aniket]
# lsof
```

- Result 🖱️



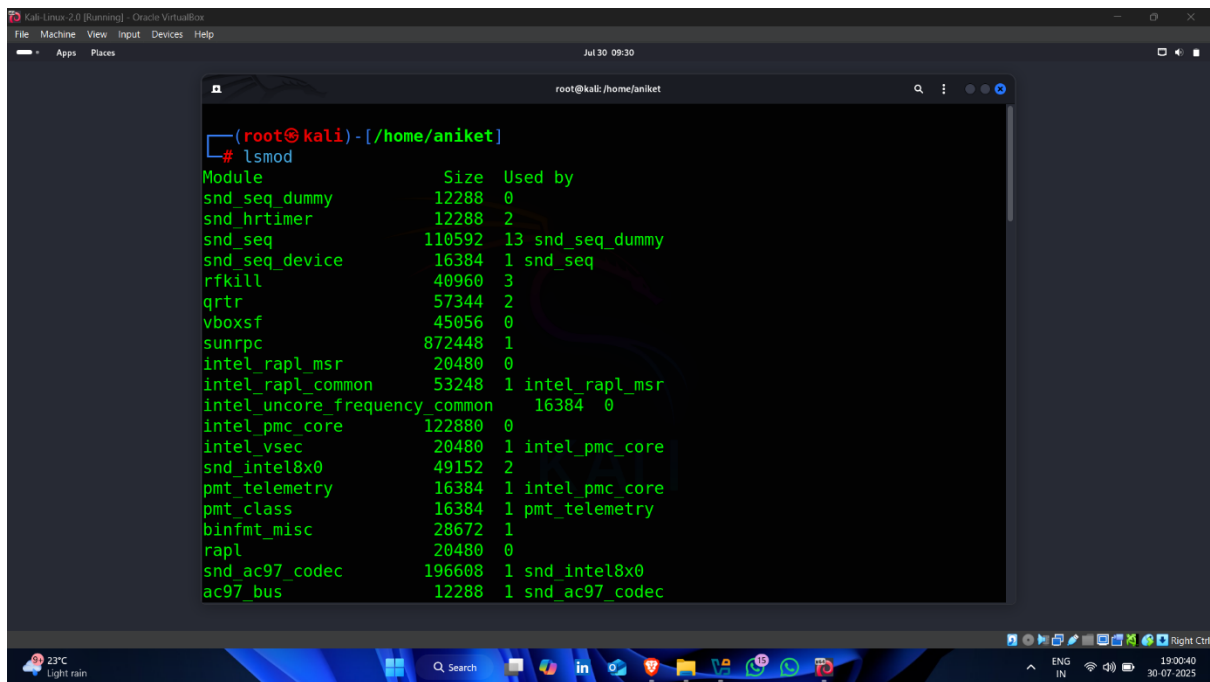
```
(root@kali) - [/home/aniket]
# lsdf
lsdf: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
Output information may be incomplete.
lsdf: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
Output information may be incomplete.
COMMAND      PID  TID TASKCMD      USER  FD  TYPE      DEVICE  SIZE
/OFF          NODE NAME
systemd      1    /      root   cwd       DIR      8,1
4096          2 /      root   rtd       DIR      8,1
systemd      1    /      root   txt       REG      8,1  10
4832      789834 /usr/lib/systemd/systemd root   mem       REG      8,1  11
3248      791294 /usr/lib/x86_64-linux-gnu/libelf-0.191.so root   mem       REG      8,1  81
systemd      1    /      root   mem       REG      8,1  570
3048      787245 /usr/lib/x86_64-linux-gnu/libzstd.so.1.5.6 root   mem       REG      8,1  42
systemd      1    /      root   mem       REG      8,1  19
6872      787276 /usr/lib/x86_64-linux-gnu/libcrypto.so.3 root   mem       REG      8,1  19
systemd      1    /      root   mem       REG      8,1  19
3952      856068 /usr/lib/x86_64-linux-gnu/libbpf.so.1.5.0 root   mem       REG      8,1  19
systemd      1    /      root   mem       REG      8,1  19
```

Command -: **lsmod** --The lsmod command in Linux is used to **list all loaded kernel modules** currently active in the system.



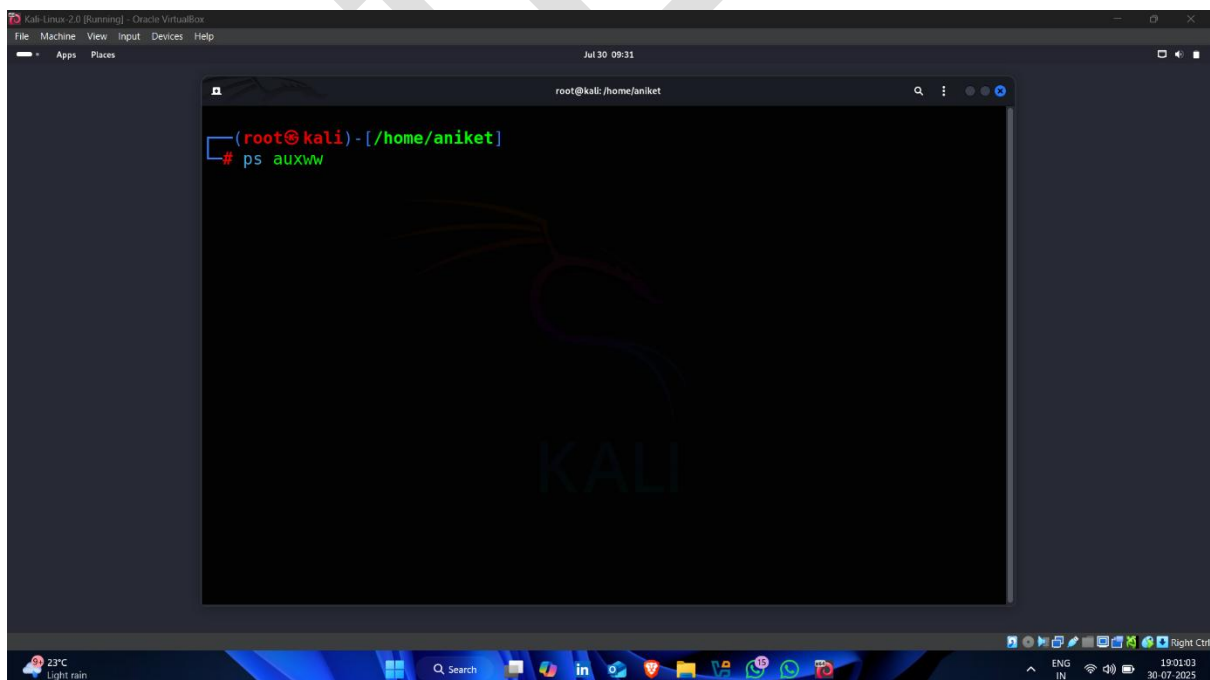
```
(root@kali) - [/home/aniket]
# lsmod
```

- Result 🖱️



```
(root@kali) - [/home/aniket]
# lsmod
Module                Size  Used by
snd_seq_dummy          12288  0
snd_hrtimer            12288  2
snd_seq               110592  13 snd_seq_dummy
snd_seq_device         16384  1 snd_seq
rfkill                 40960  3
qrtr                   57344  2
vboxsf                 45056  0
sunrpc                 872448  1
intel_rapl_msr         20480  0
intel_rapl_common      53248  1 intel_rapl_msr
intel_uncore_frequency_common 16384  0
intel_pmc_core         122880  0
intel_vsec             20480  1 intel_pmc_core
snd_intel8x0           49152  2
pmt_telemetry          16384  1 intel_pmc_core
pmt_class              16384  1 pmt_telemetry
binfmt_misc            28672  1
rapl                   20480  0
snd_ac97_codec         196608  1 snd_intel8x0
ac97_bus               12288  1 snd_ac97_codec
```

Command -: `ps auxww` -- is used in Linux to display **detailed information about all running processes**, with **full command-line arguments**, even if they are very long.



```
(root@kali) - [/home/aniket]
# ps auxww
```

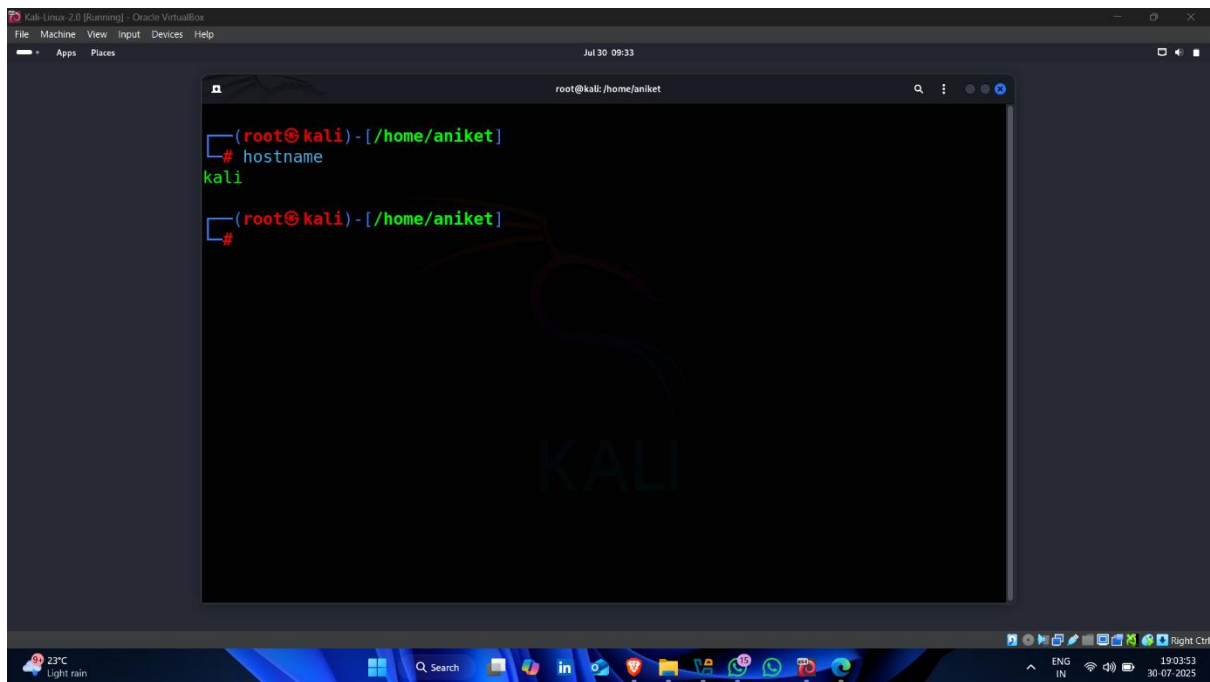
- Result 🙌

```
(root@kali) - [/home/aniket]
# ps auxww
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.3  0.4  22456 13832 ?        Ss   09:25   0:01 /sbin/init sp
lash
root           2  0.0  0.0      0     0 ?        S    09:25   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    09:25   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   09:25   0:00 [kworker/R-kvfree_rcu_reclaim]
root           5  0.0  0.0      0     0 ?        I<   09:25   0:00 [kworker/R-rcu_gp]
root           6  0.0  0.0      0     0 ?        I<   09:25   0:00 [kworker/R-synct_wq]
root           7  0.0  0.0      0     0 ?        I<   09:25   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   09:25   0:00 [kworker/R-netns]
root           9  0.0  0.0      0     0 ?        I    09:25   0:00 [kworker/0:0-events]
root          10  0.1  0.0      0     0 ?        I    09:25   0:00 [kworker/0:1-events]
root          12  0.0  0.0      0     0 ?        I    09:25   0:00 [kworker/u8:0]
```

Command -: `hostname --` is used in Linux to **display the system's hostname**, which is the **name assigned to the computer** on a network.

```
(root@kali) - [/home/aniket]
# hostname
kali
```

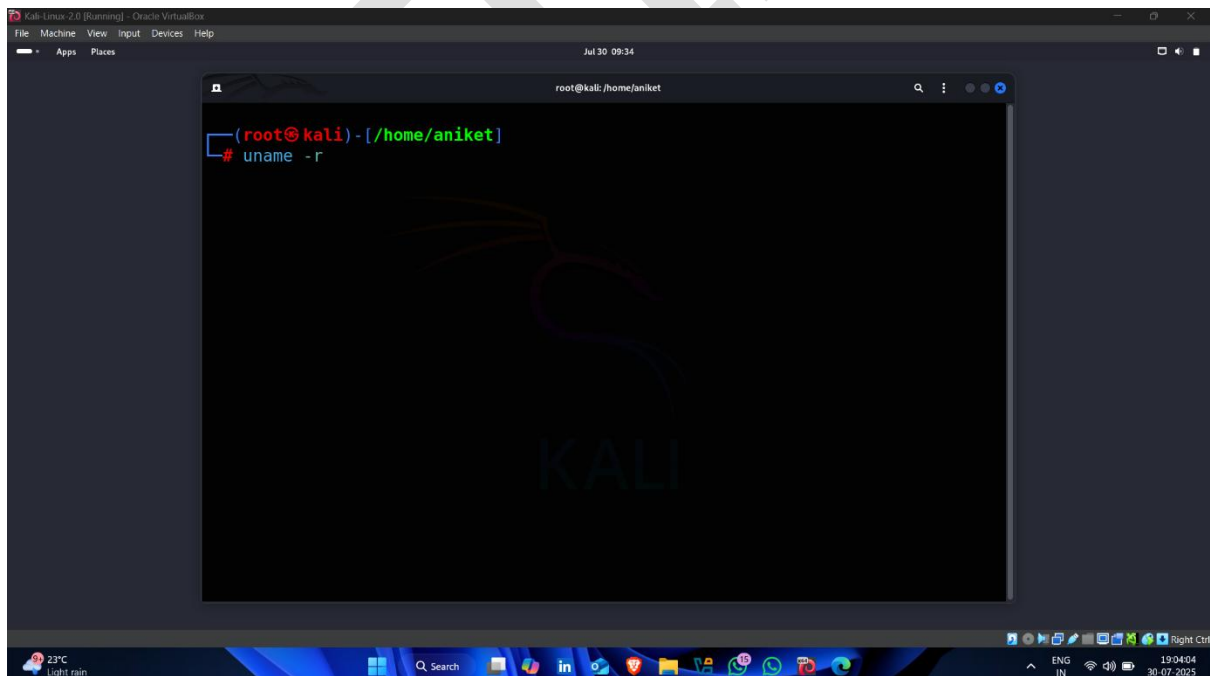

- Result 🖱️



A screenshot of a Kali Linux terminal window running inside an Oracle VM VirtualBox. The terminal title bar shows 'root@kali: /home/aniket'. The prompt is '(root@kali) - [/home/aniket]'. The user has entered the command 'hostname', and the output is 'kali'. The terminal background features a large, faint 'KALI' logo. The host desktop at the bottom shows a taskbar with various application icons, a search bar, and system status indicators including temperature (23°C), weather (Light rain), and time (19:03:53 on 30-07-2025).

```
(root@kali) - [/home/aniket]
# hostname
kali
(root@kali) - [/home/aniket]
#
```

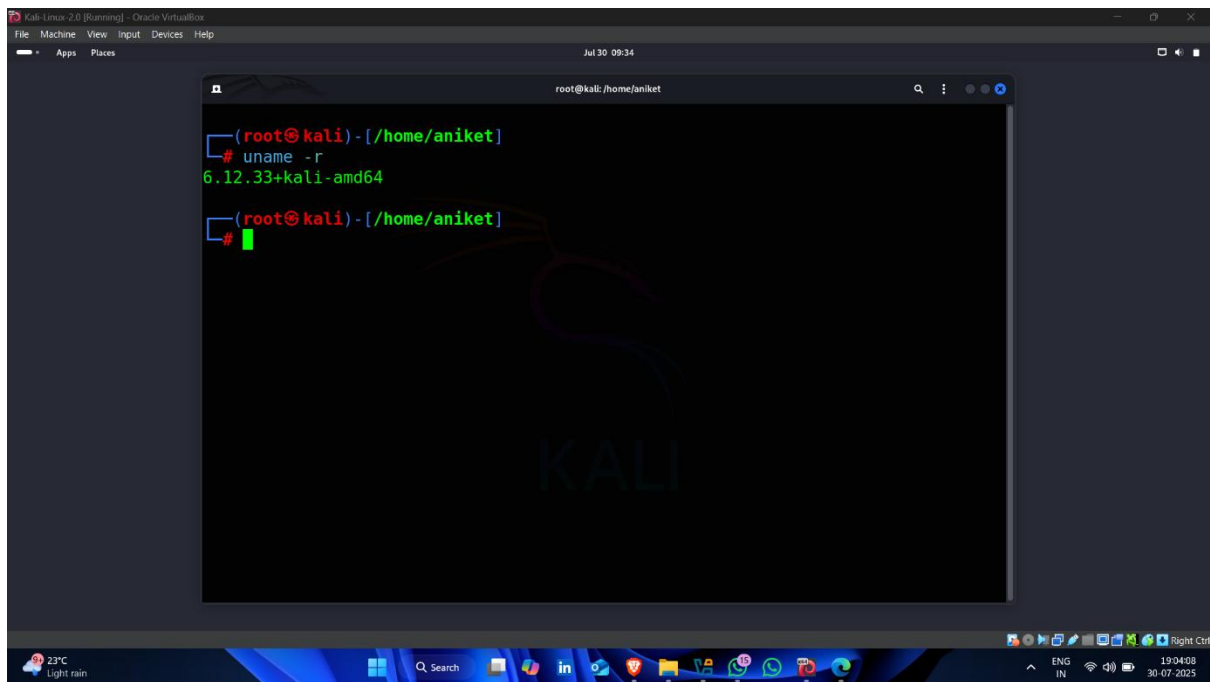
Command -: `uname -r` -- is used in Linux to **display the kernel version** currently running on the system.



A screenshot of a Kali Linux terminal window, similar to the one above. The terminal title bar shows 'root@kali: /home/aniket'. The prompt is '(root@kali) - [/home/aniket]'. The user has entered the command 'uname -r'. The terminal background features a large, faint 'KALI' logo. The host desktop at the bottom shows a taskbar with various application icons, a search bar, and system status indicators including temperature (23°C), weather (Light rain), and time (19:04:04 on 30-07-2025).

```
(root@kali) - [/home/aniket]
# uname -r
```

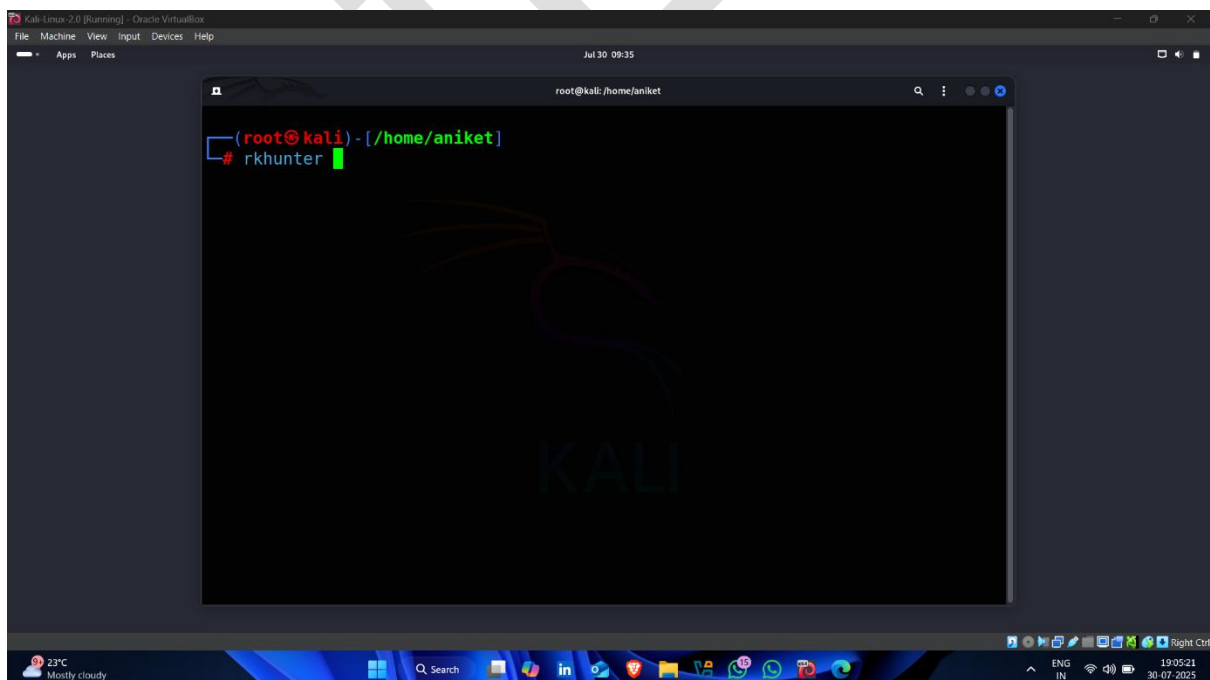
- Result 📍



A screenshot of a Kali Linux terminal window running inside an Oracle VM VirtualBox. The terminal shows the command `uname -r` being executed, resulting in the output `6.12.33+kali-amd64`. The terminal window has a title bar that reads "root@kali: /home/aniket". The background of the terminal window features a large, stylized "KALI" logo. The desktop environment is visible at the bottom, showing a taskbar with various application icons and a system tray with a clock and weather widget.

```
(root@kali) - [/home/aniket]
# uname -r
6.12.33+kali-amd64
(root@kali) - [/home/aniket]
#
```

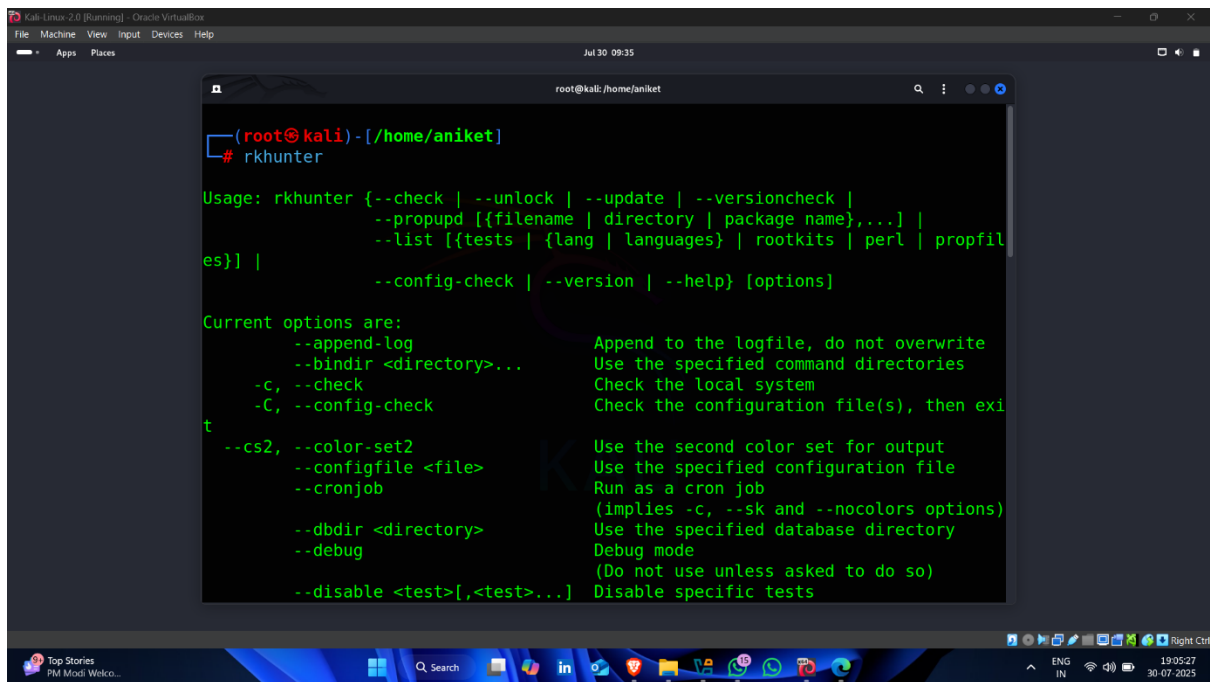
Command -: **rkhunter** -- rkhunter stands for **Rootkit Hunter** — it's an open-source **command-line tool** in Linux used to **detect known rootkits, backdoors, and possible local exploits** on a system.



A screenshot of a Kali Linux terminal window running inside an Oracle VM VirtualBox. The terminal shows the command `rkhunter` being executed. The terminal window has a title bar that reads "root@kali: /home/aniket". The background of the terminal window features a large, stylized "KALI" logo. The desktop environment is visible at the bottom, showing a taskbar with various application icons and a system tray with a clock and weather widget.

```
(root@kali) - [/home/aniket]
# rkhunter
```

- Detailed About Rkhunter 🖱️

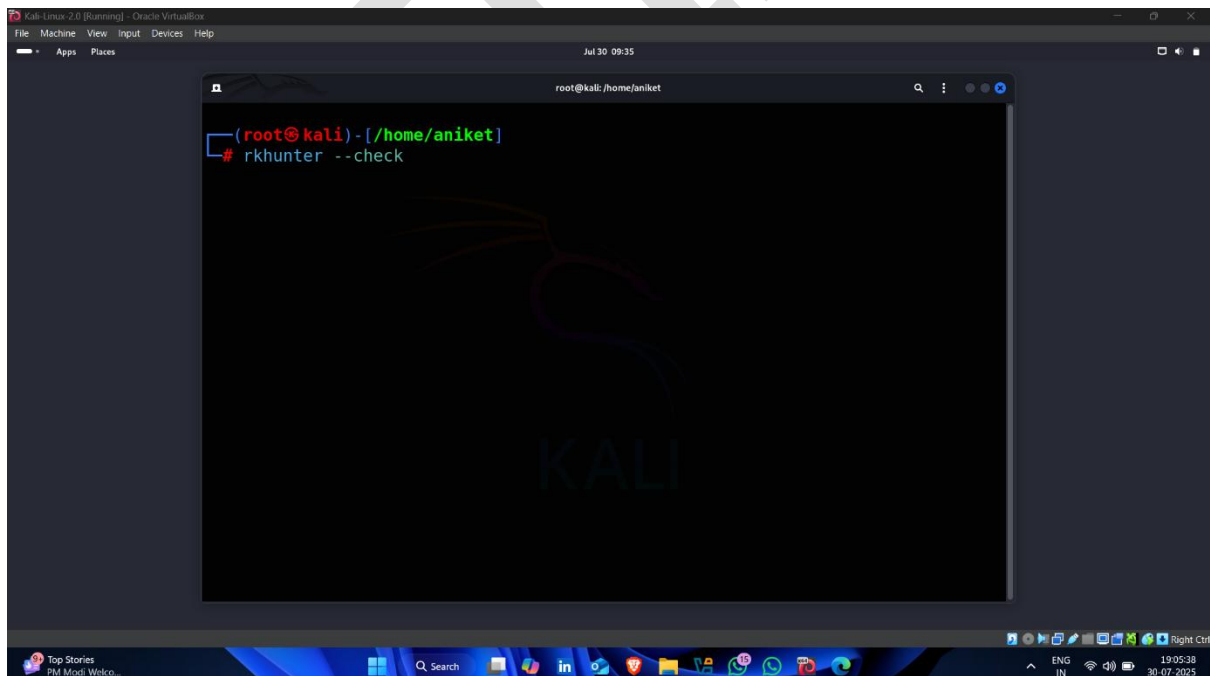


```
(root@kali) - [/home/aniket]
# rkhunter

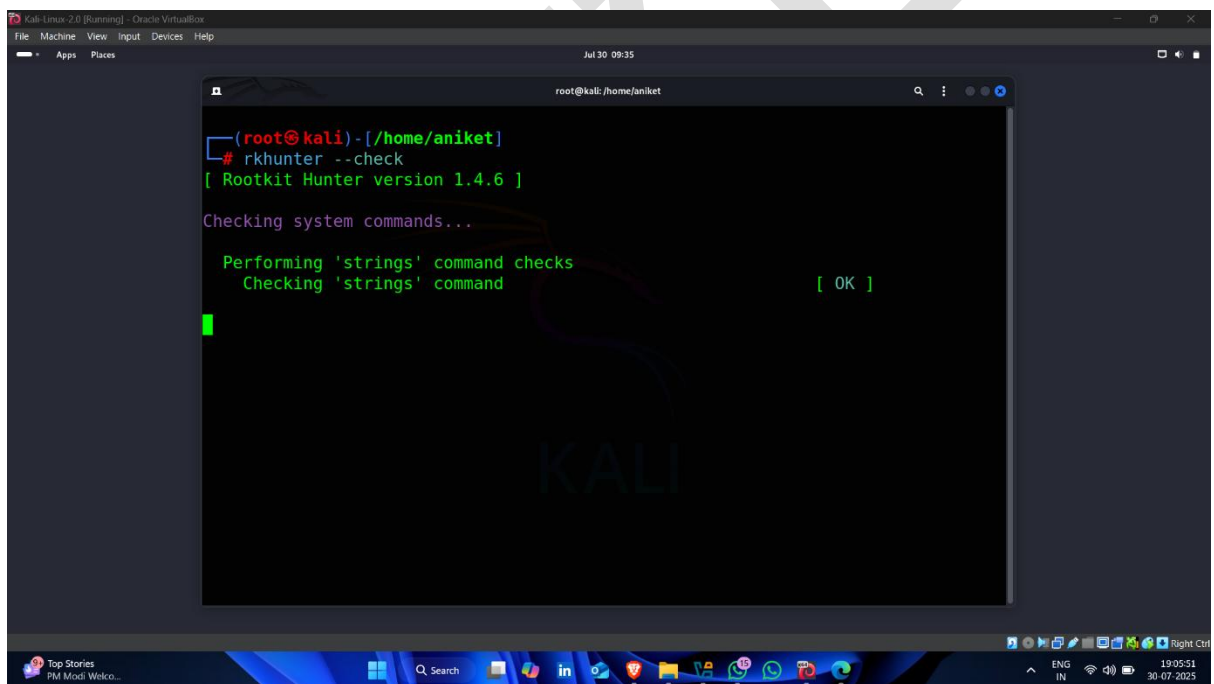
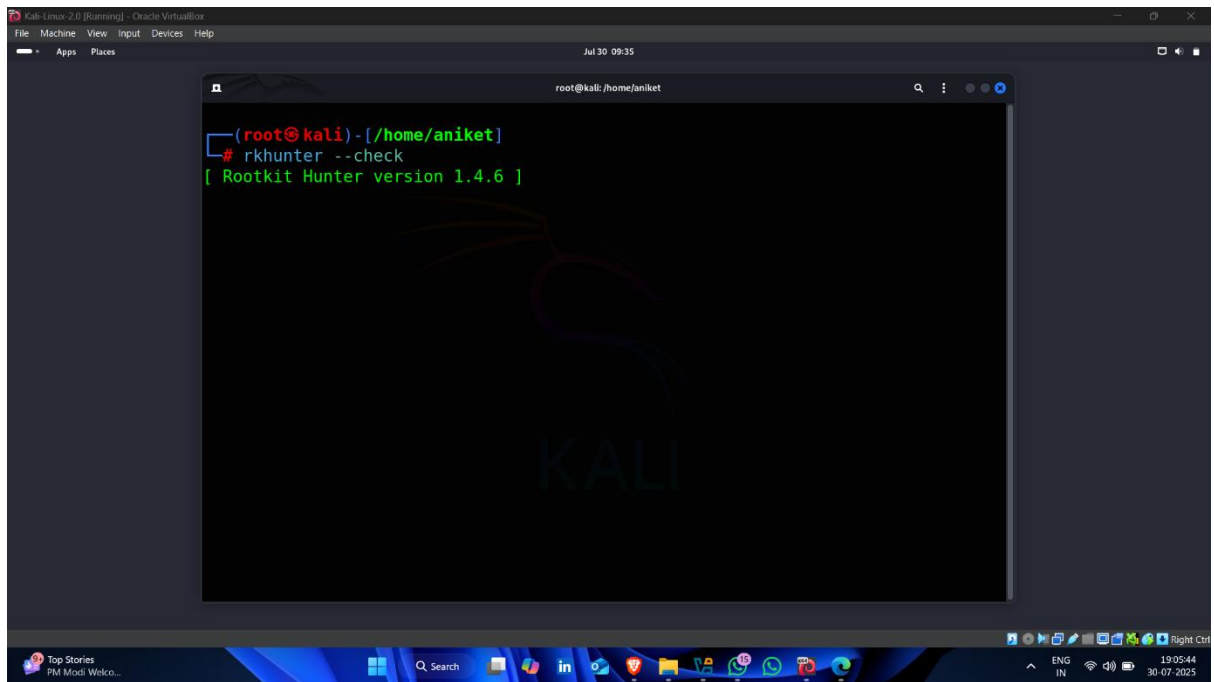
Usage: rkhunter [--check | --unlock | --update | --versioncheck |
               --propupd [{filename | directory | package name},...] |
               --list [{tests | {lang | languages} | rootkits | perl | propfil
es}] |
               --config-check | --version | --help] [options]

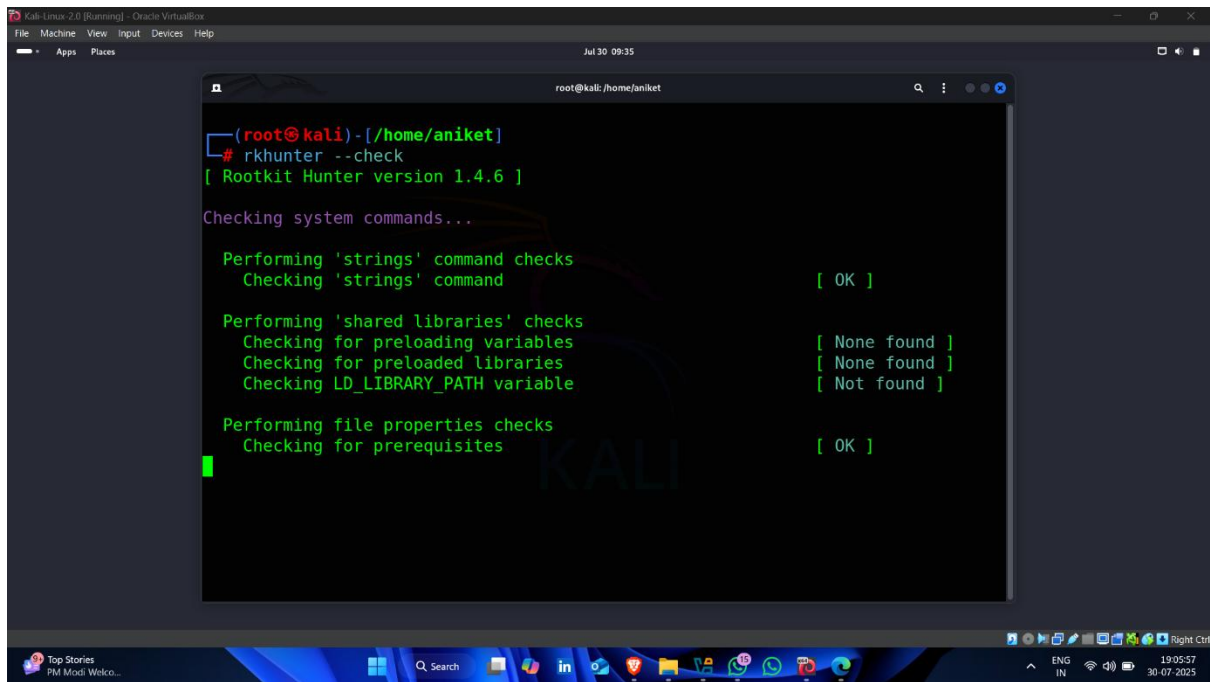
Current options are:
--append-log           Append to the logfile, do not overwrite
--bindir <directory>... Use the specified command directories
-c, --check            Check the local system
-C, --config-check     Check the configuration file(s), then exit
--cs2, --color-set2    Use the second color set for output
--configfile <file>    Use the specified configuration file
--cronjob              Run as a cron job
                       (implies -c, --sk and --nocolors options)
--dbdir <directory>    Use the specified database directory
--debug               Debug mode
                       (Do not use unless asked to do so)
--disable <test>[,<test>...] Disable specific tests
```

Command -: **rkhunter --check** -- it initiates a **rootkit scan** on your Linux system. Here's what it does and how to interpret it:



```
(root@kali) - [/home/aniket]
# rkhunter --check
```





```
(root@kali) - [/home/aniket]
# rkhunter --check
[ Rootkit Hunter version 1.4.6 ]

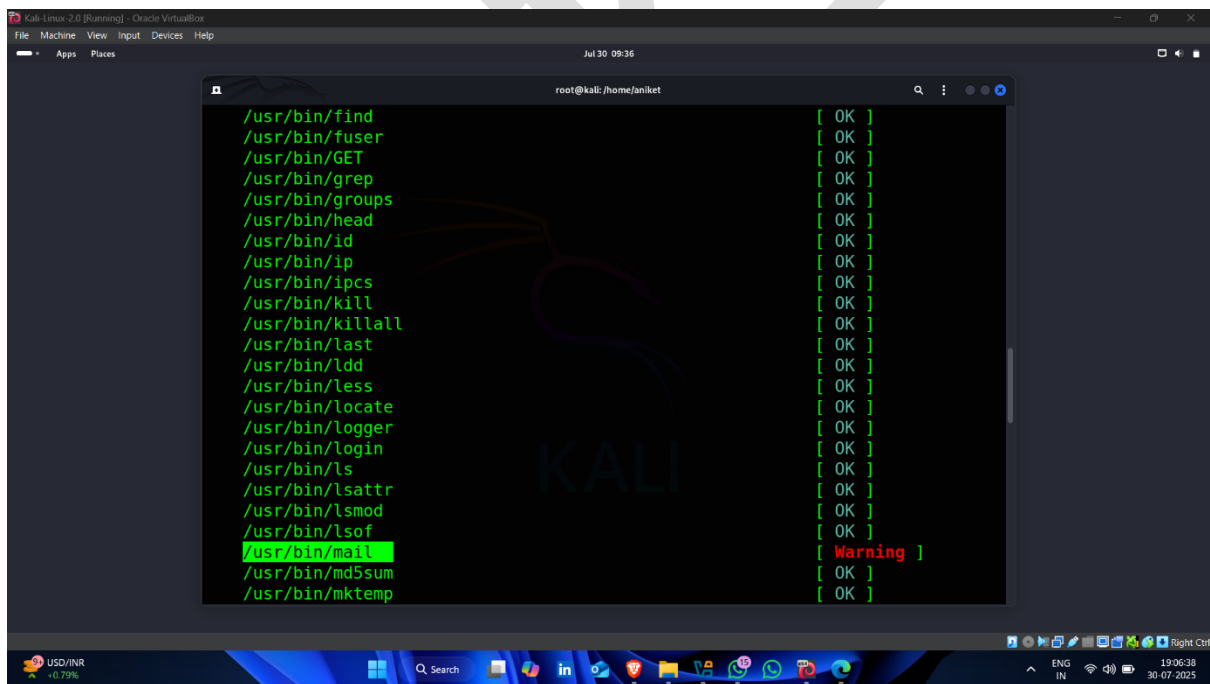
Checking system commands...

Performing 'strings' command checks
Checking 'strings' command [ OK ]

Performing 'shared libraries' checks
Checking for preloading variables [ None found ]
Checking for preloaded libraries [ None found ]
Checking LD_LIBRARY_PATH variable [ Not found ]

Performing file properties checks
Checking for prerequisites [ OK ]
```

• Result 📌 ✓



```
/usr/bin/find [ OK ]
/usr/bin/fuser [ OK ]
/usr/bin/GET [ OK ]
/usr/bin/grep [ OK ]
/usr/bin/groups [ OK ]
/usr/bin/head [ OK ]
/usr/bin/id [ OK ]
/usr/bin/ip [ OK ]
/usr/bin/ipcs [ OK ]
/usr/bin/kill [ OK ]
/usr/bin/killall [ OK ]
/usr/bin/last [ OK ]
/usr/bin/ldd [ OK ]
/usr/bin/less [ OK ]
/usr/bin/locate [ OK ]
/usr/bin/logger [ OK ]
/usr/bin/login [ OK ]
/usr/bin/ls [ OK ]
/usr/bin/lsattr [ OK ]
/usr/bin/lsmmod [ OK ]
/usr/bin/lsof [ OK ]
/usr/bin/mail [ Warning ]
/usr/bin/md5sum [ OK ]
/usr/bin/mktemp [ OK ]
```

Lab-2-: Investigate Forensics Image of Linux System

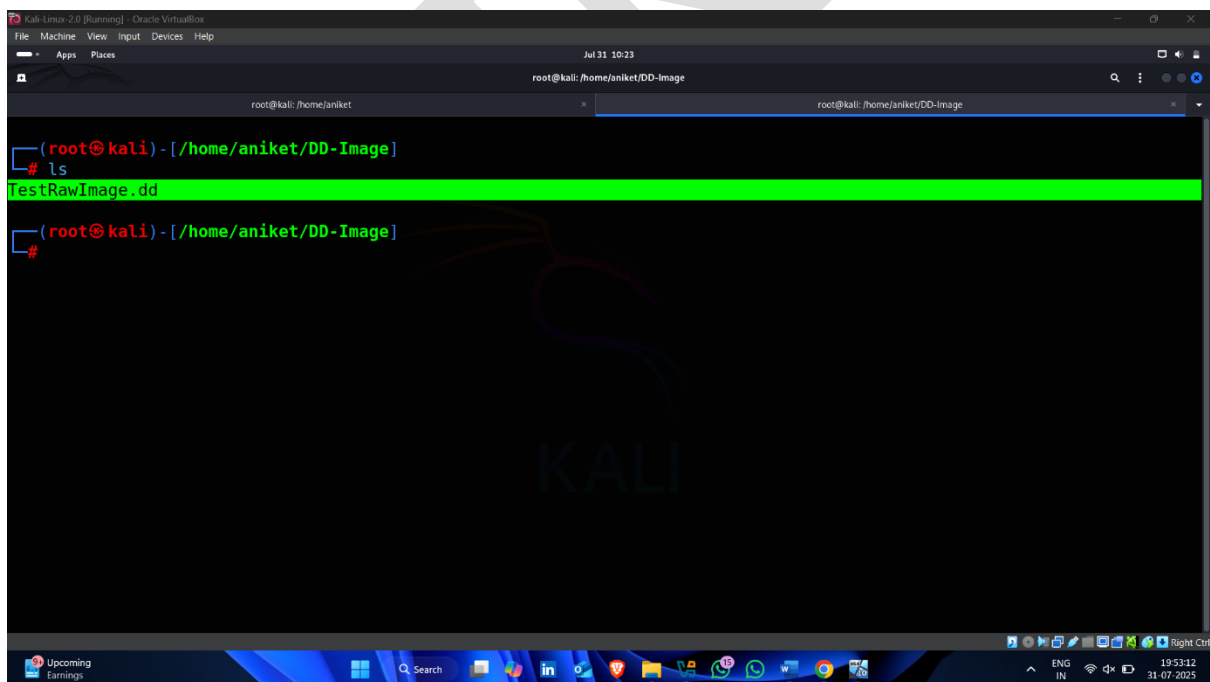
Investigating a forensics image of a Linux system means analyzing a **bit-by-bit copy** (also called a **disk image**, e.g., .dd, .img, .E01) of a Linux machine to **uncover evidence**

A) Investigate Forensics Image of Linux System Using Autopsy

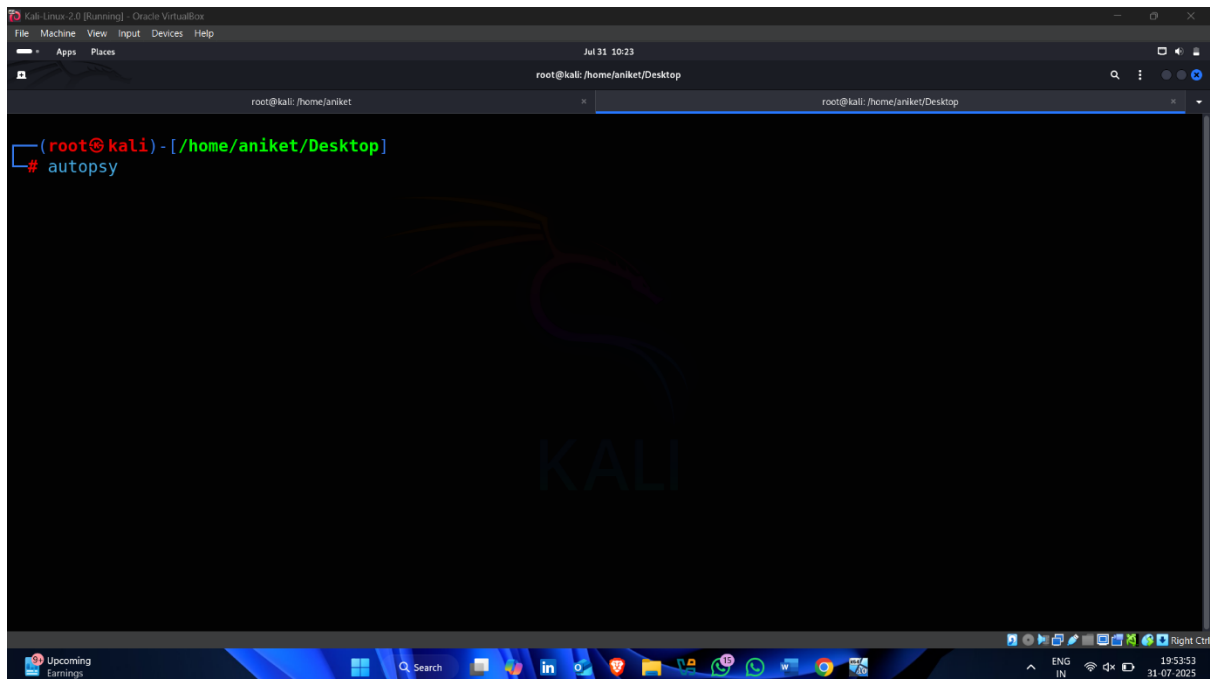
Autopsy is a **GUI-based digital forensics platform** used to **analyze hard drive images**, memory dumps, and other digital evidence. It's built on **The Sleuth Kit (TSK)** and is widely used in **digital forensics, incident response, and law enforcement investigations**.

How to use it -:

- I already have image file 🖱️

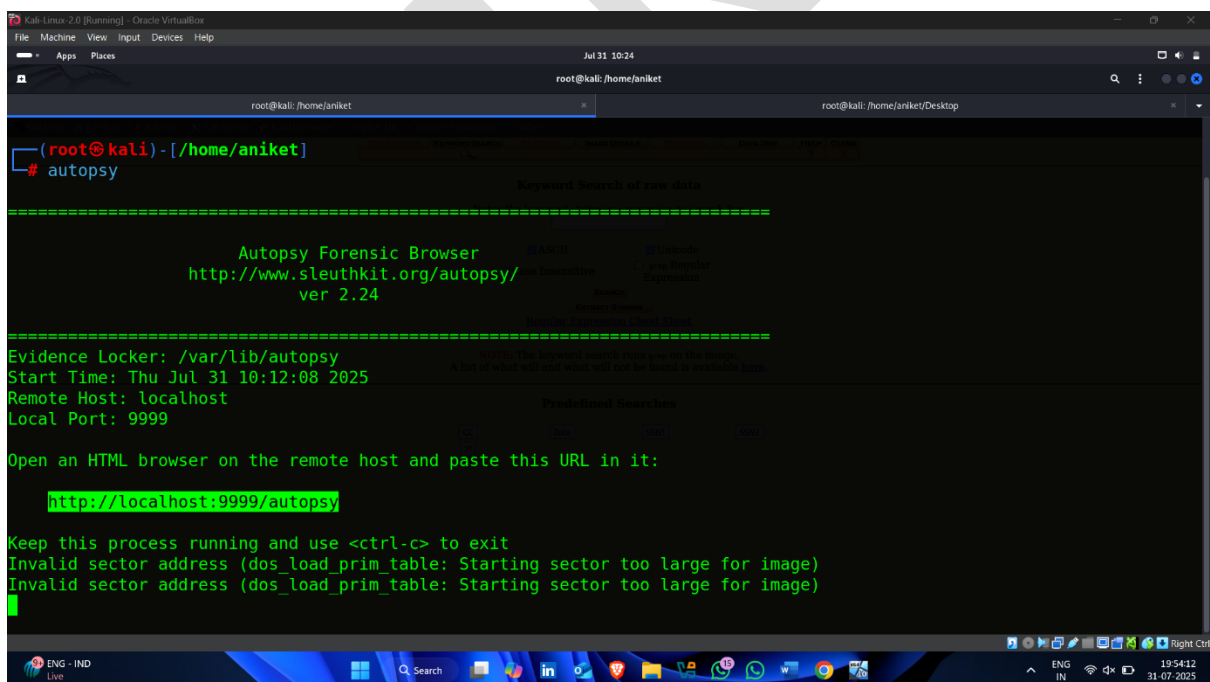


- **Type autopsy**



The screenshot shows a Kali Linux terminal window with the prompt `(root@kali) - [/home/aniket/Desktop]`. The command `# autopsy` has been entered. The terminal background features a large, stylized 'KALI' logo.

- **Copy this link and paste on browser**



The screenshot shows the output of the `autopsy` command. It displays the 'Autopsy Forensic Browser' version 2.24, the evidence locker path `/var/lib/autopsy`, the start time `Thu Jul 31 10:12:08 2025`, and the remote host `localhost`. It also provides a URL to open in a browser: `http://localhost:9999/autopsy`. The output includes a warning about an invalid sector address.

```
(root@kali) - [/home/aniket]
# autopsy

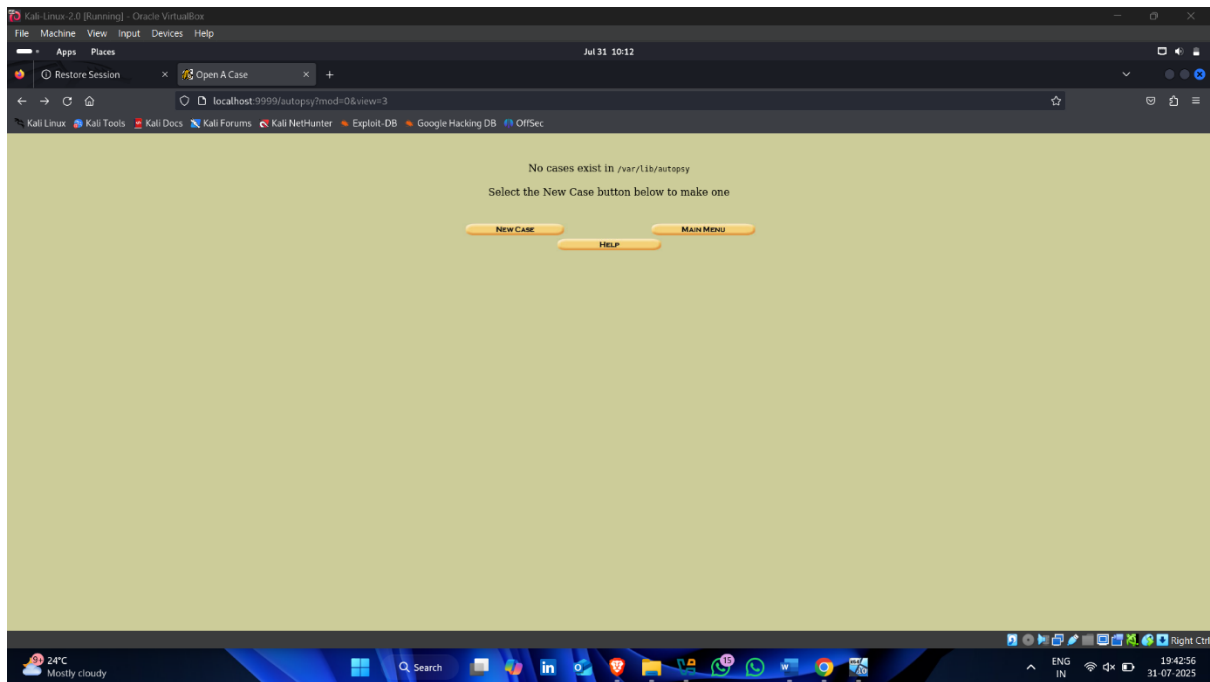
Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24

Evidence Locker: /var/lib/autopsy
Start Time: Thu Jul 31 10:12:08 2025
Remote Host: localhost
Local Port: 9999

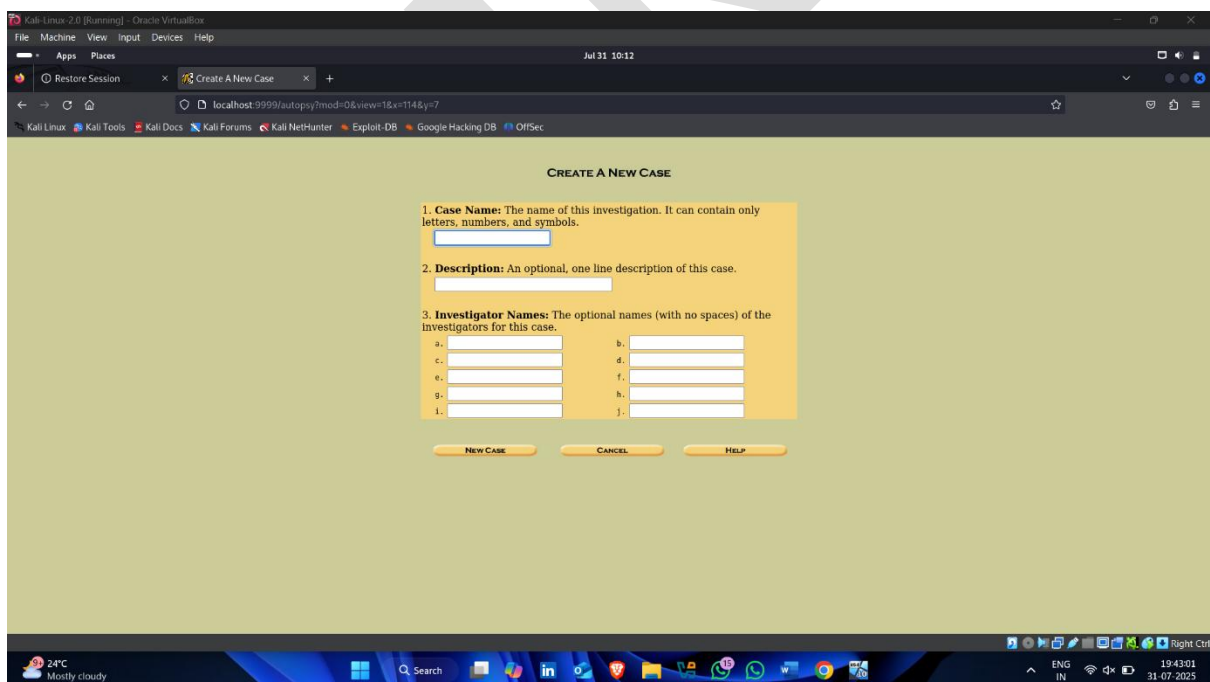
Open an HTML browser on the remote host and paste this URL in it:
http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
Invalid sector address (dos_load_prim_table: Starting sector too large for image)
Invalid sector address (dos_load_prim_table: Starting sector too large for image)
```

- Now click on **new case**



- Enter case name



- Enter case **Description** and click on **new case**

The screenshot shows a web browser window with the title 'Kali Linux 2.0 [Running] - Oracle VM VirtualBox'. The address bar shows 'localhost:9999/autopsy?mod=0&view=1&x=114&y=7'. The page content is titled 'CREATE A NEW CASE' and contains three sections:

- 1. Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols. The input field contains '101-Aniket'.
- 2. Description:** An optional, one line description of this case. The input field contains 'Linux-Image'.
- 3. Investigator Names:** The optional names (with no spaces) of the Investigators for this case. There are two columns of input fields labeled a. through j.

At the bottom of the form are three buttons: 'NEW CASE', 'CANCEL', and 'HELP'.

- Click on **Add Host**

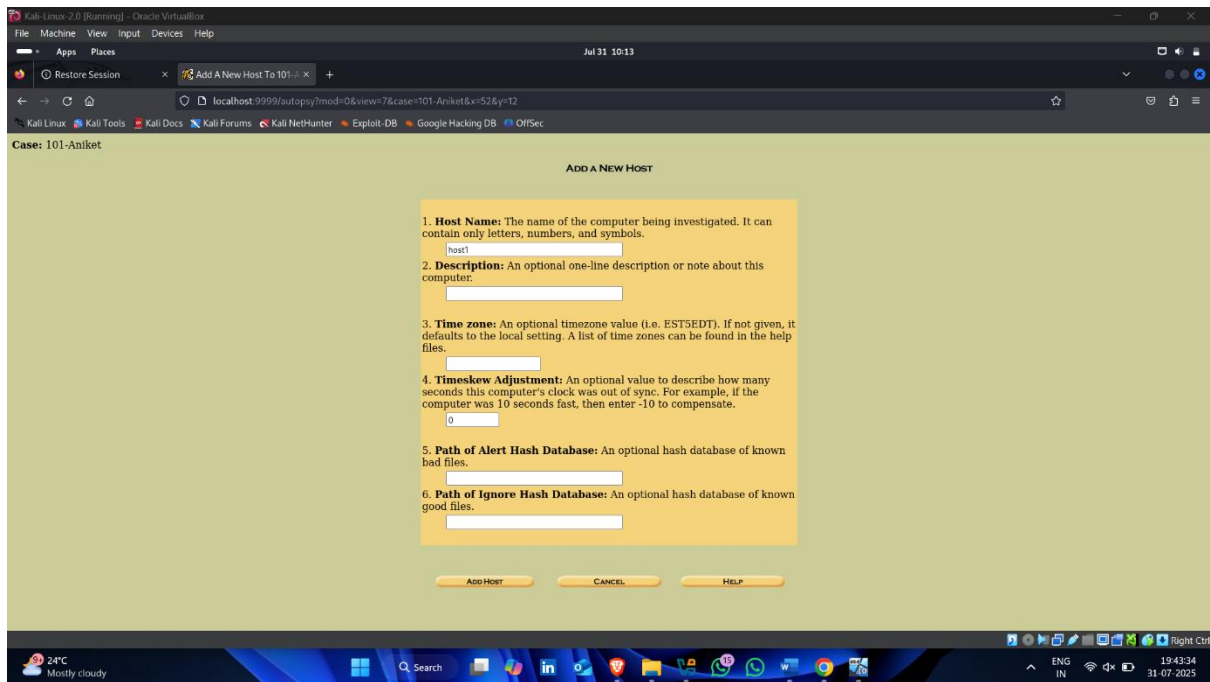
The screenshot shows the same web browser window, but the page content has changed to 'Creating Case: 101-Aniket'. It displays the following text:

Case directory (/var/lib/autopsy/101-Aniket/) created
Configuration file (/var/lib/autopsy/101-Aniket/case.conf) created

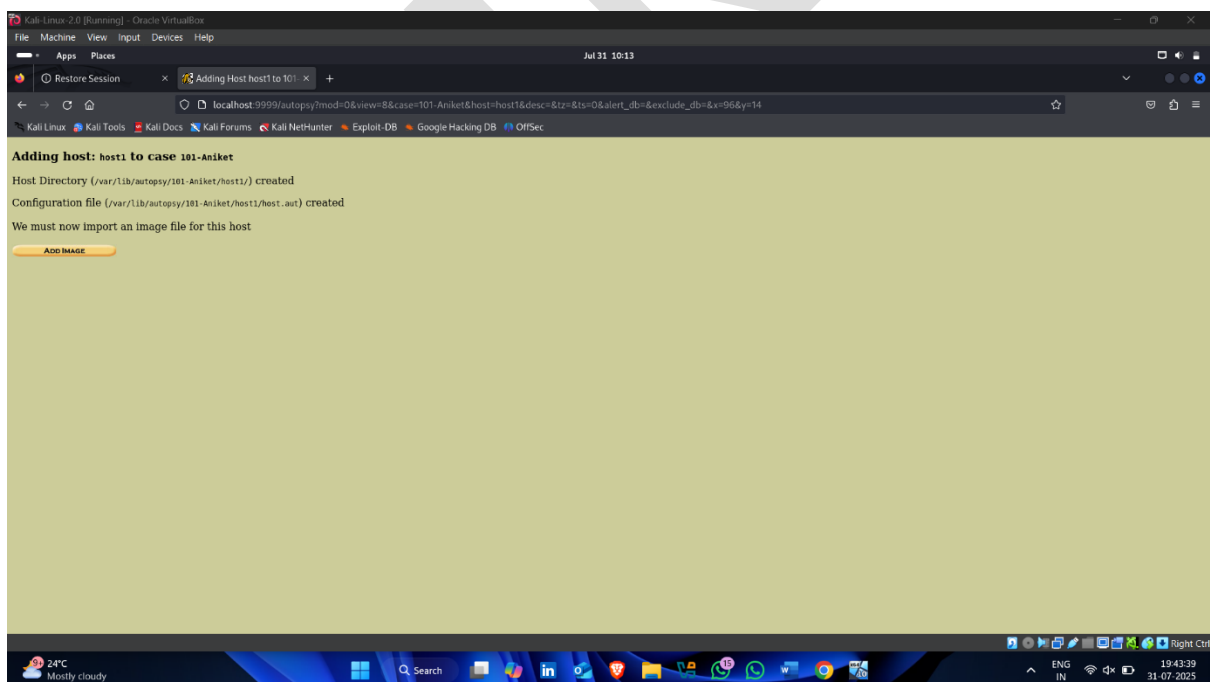
We must now create a host for this case.

Below this text is a single button labeled 'ADD HOST'.

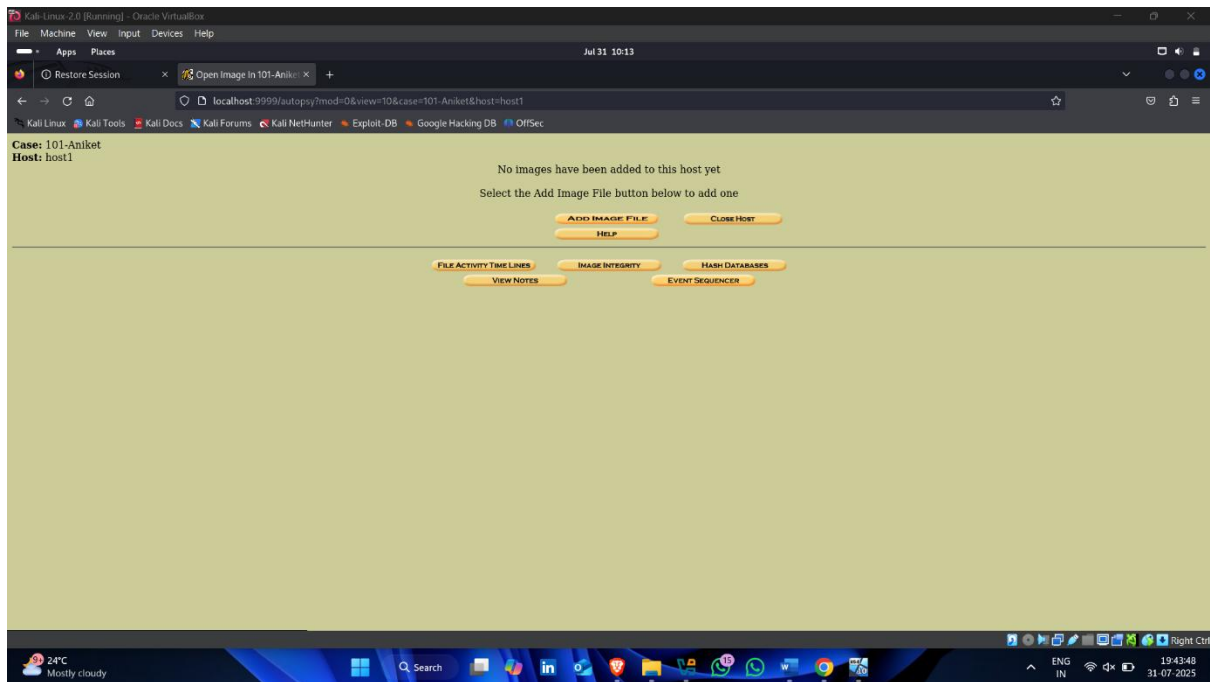
- Enter **host name and other details** and click on **Add Host**



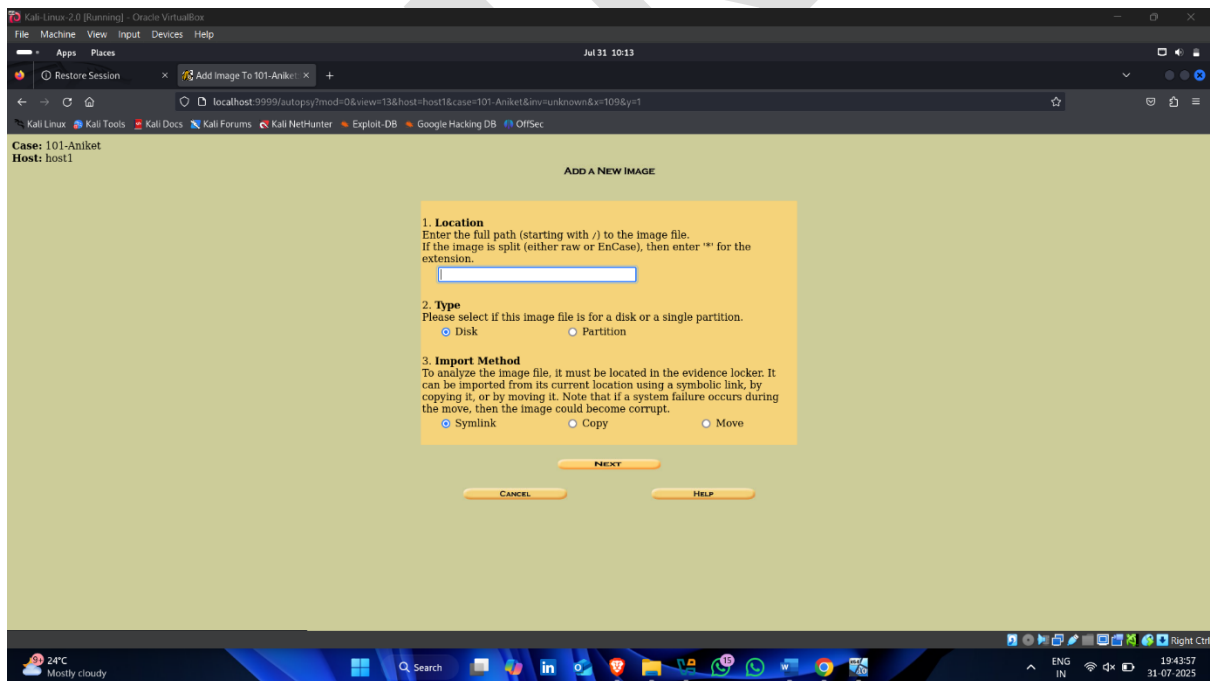
- Click on **Add Image**



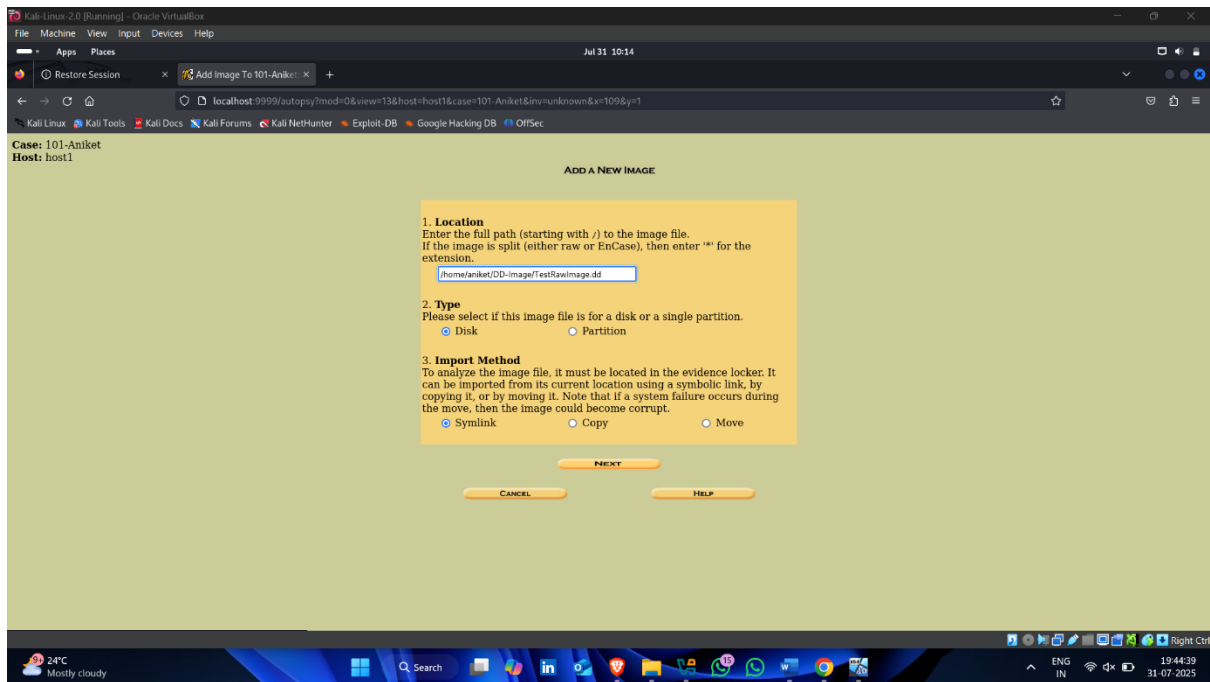
- Click on **Add Image File**



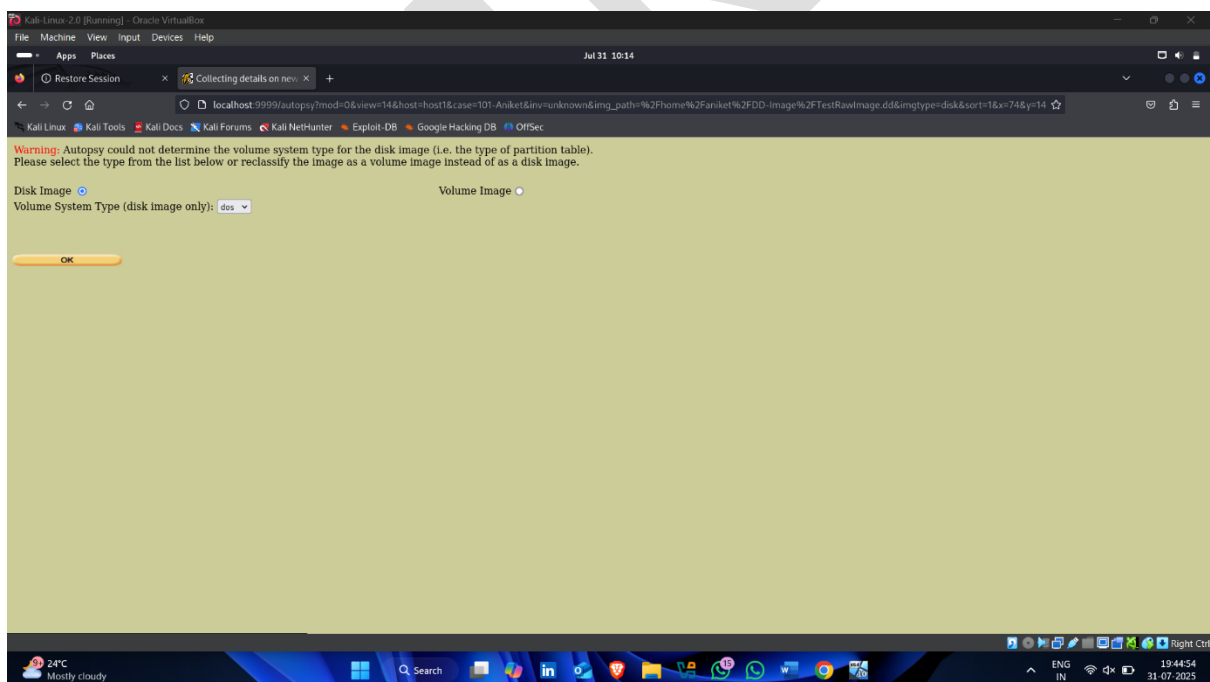
- Enter path of **image file**



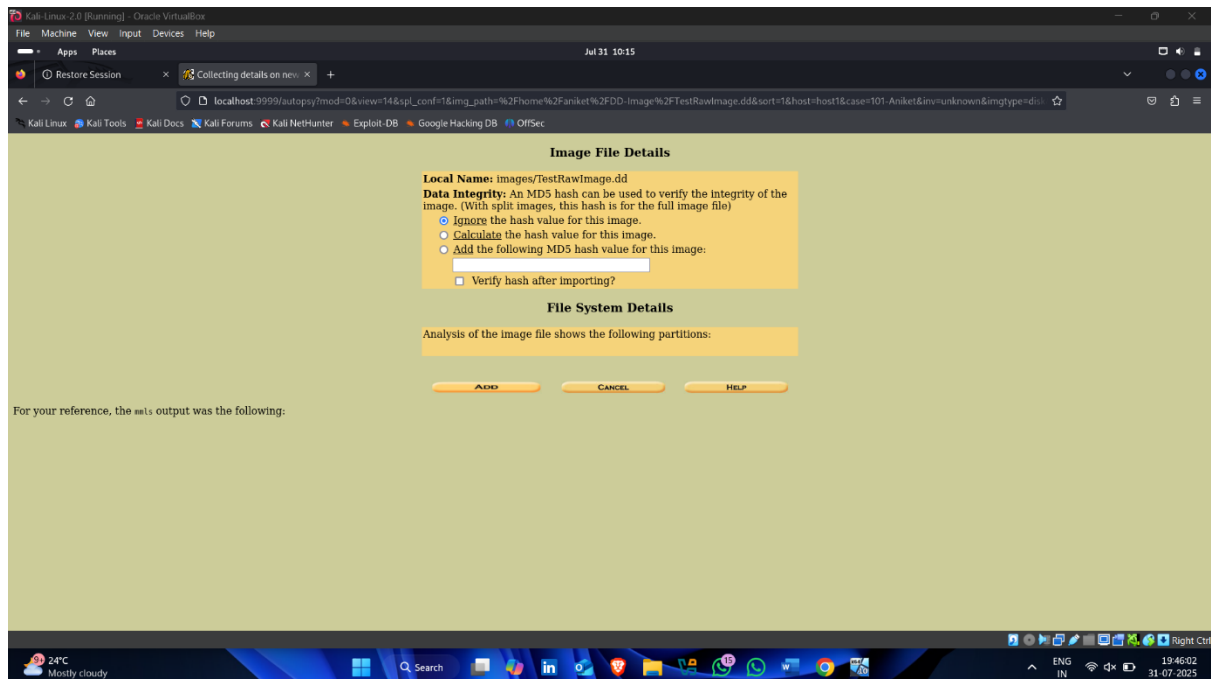
- Click on **next**



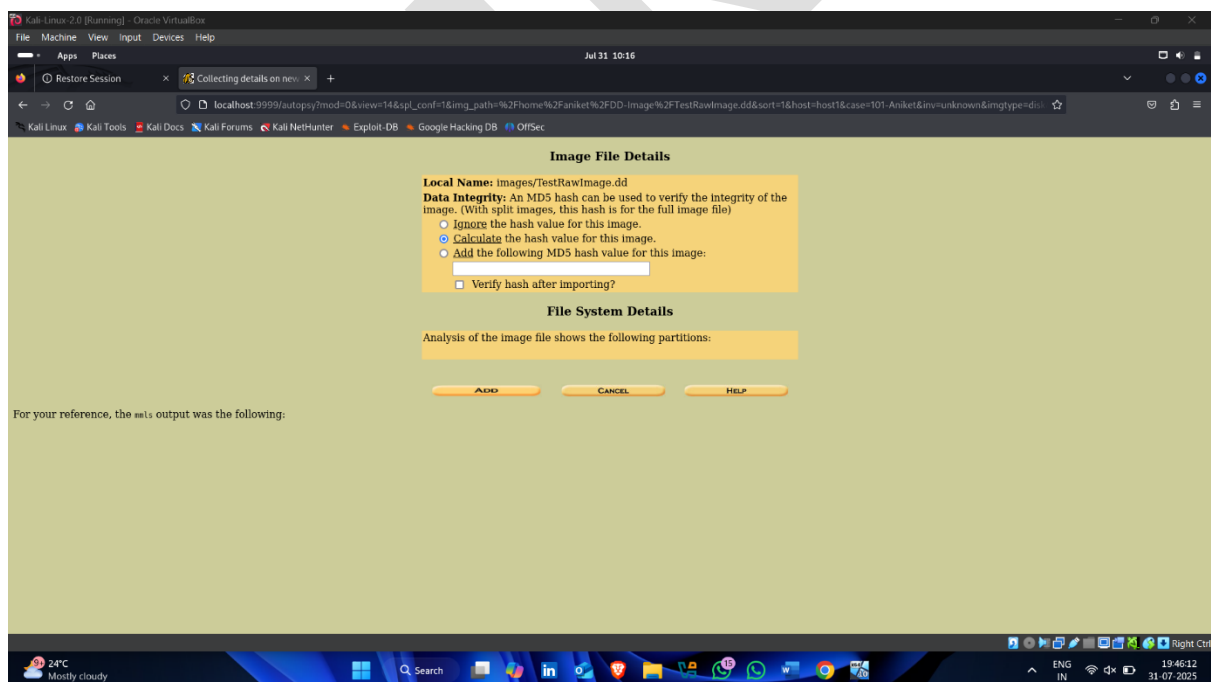
- Click on **ok**



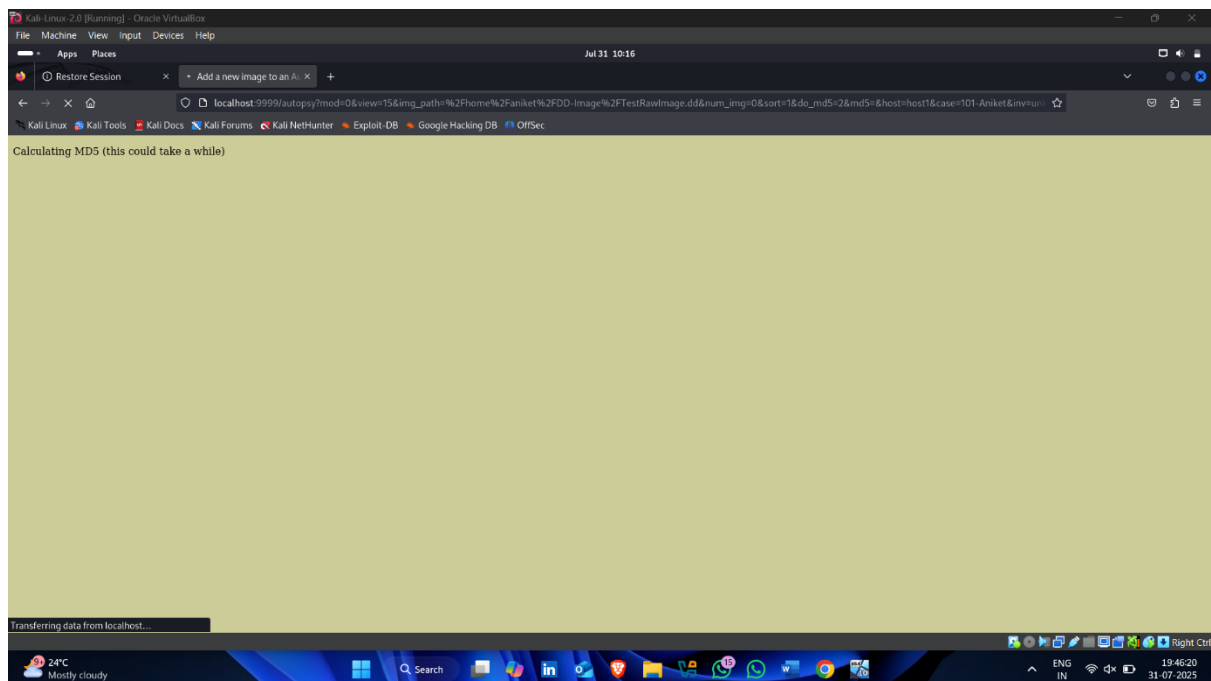
- Now select Second Option



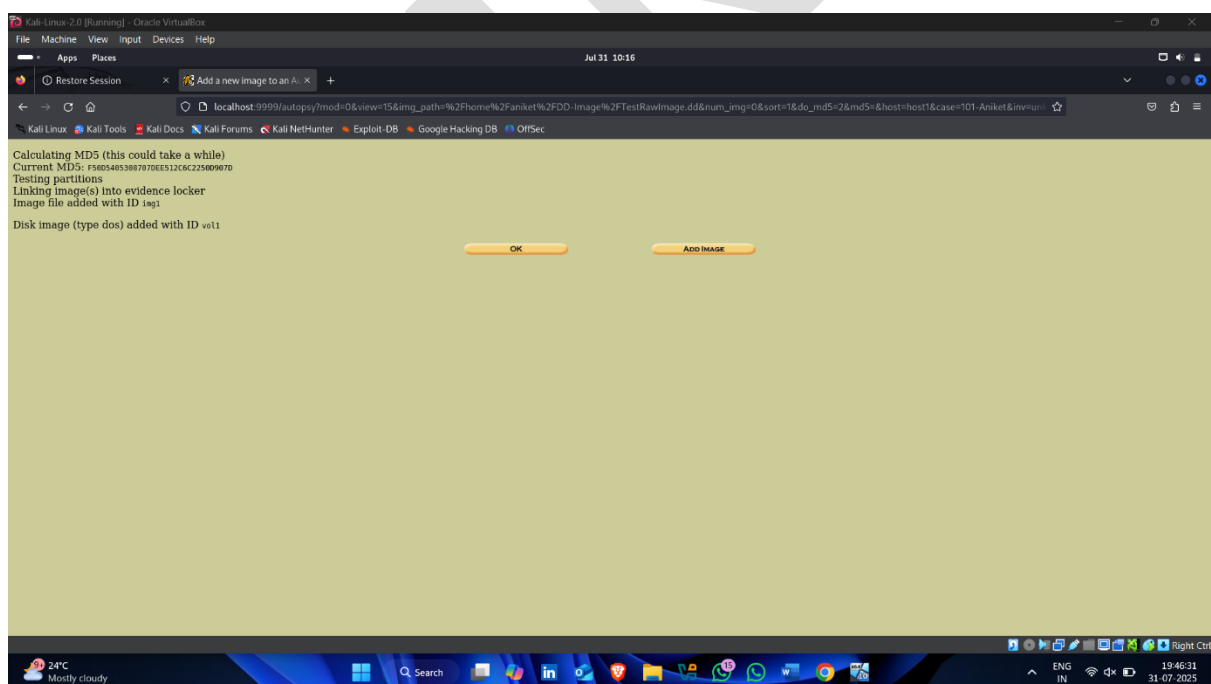
- Click on **Add**



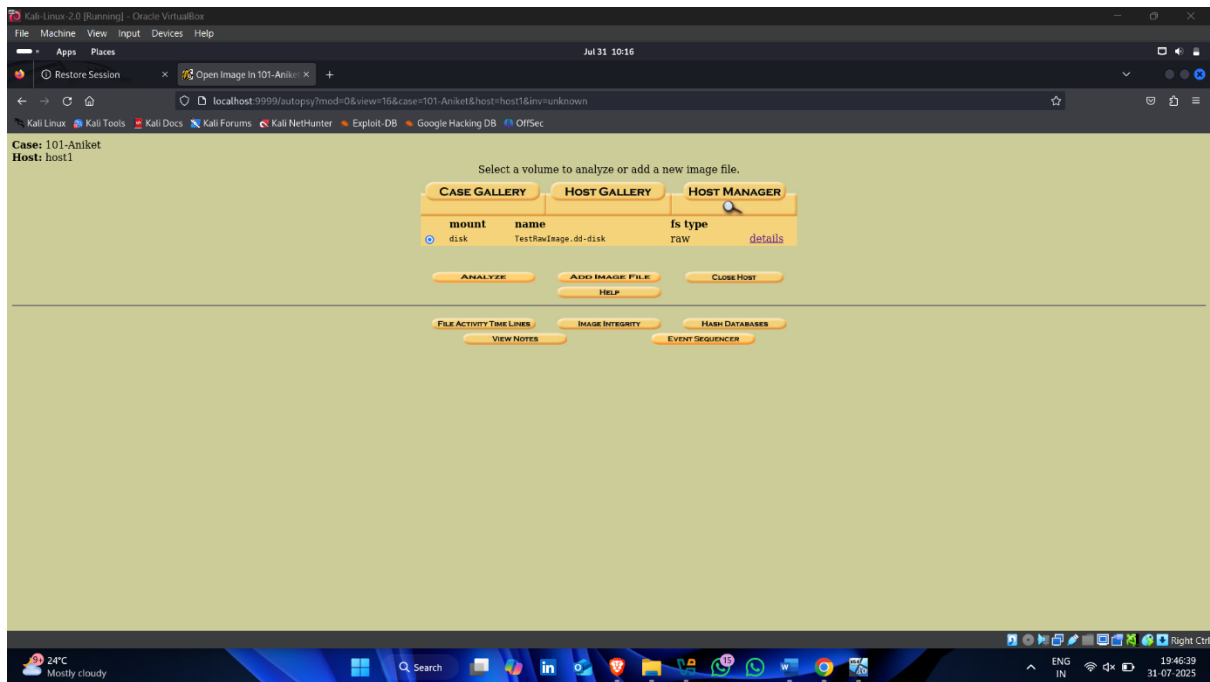
- Autopsy Working Started



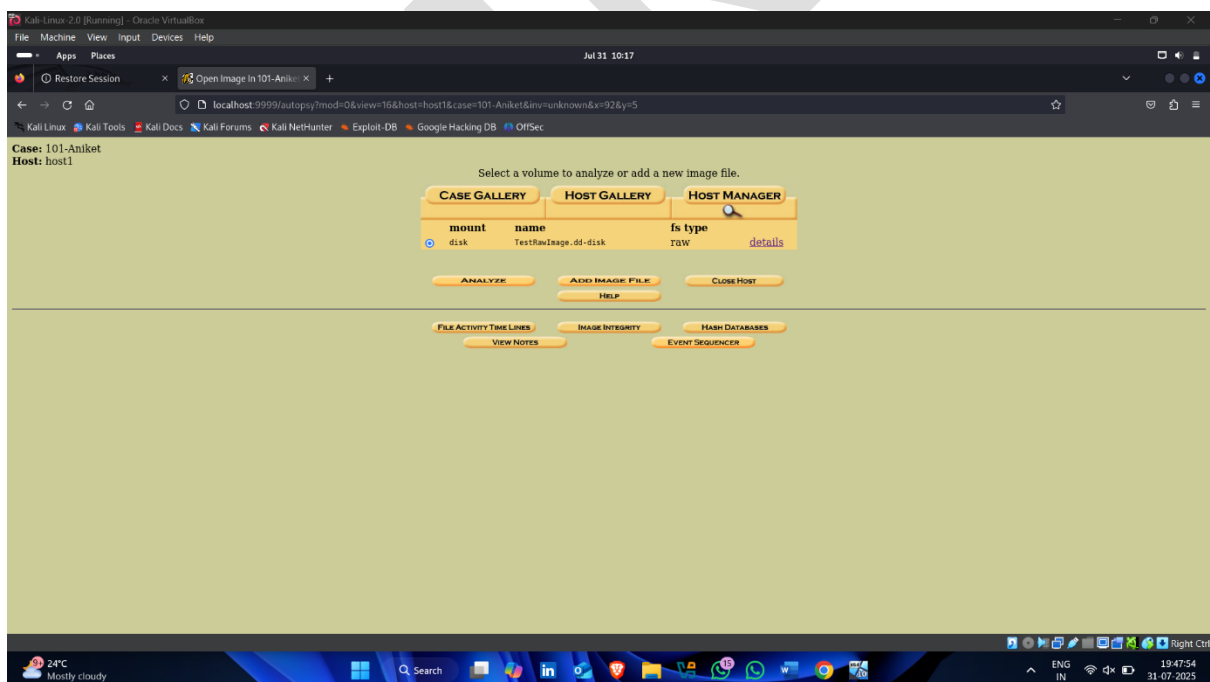
- Click on **Ok**

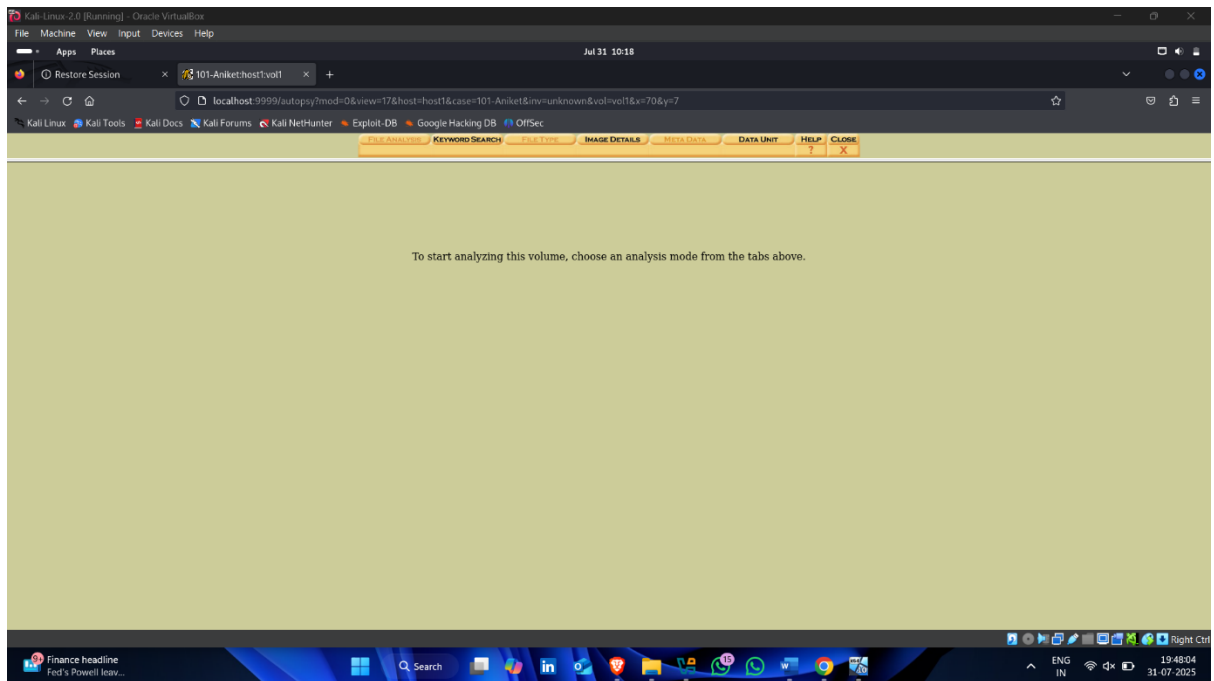


- Image mounted



- Click on Analyze





Lab-3-: Investigate Forensics Image of Linux Memory Dump

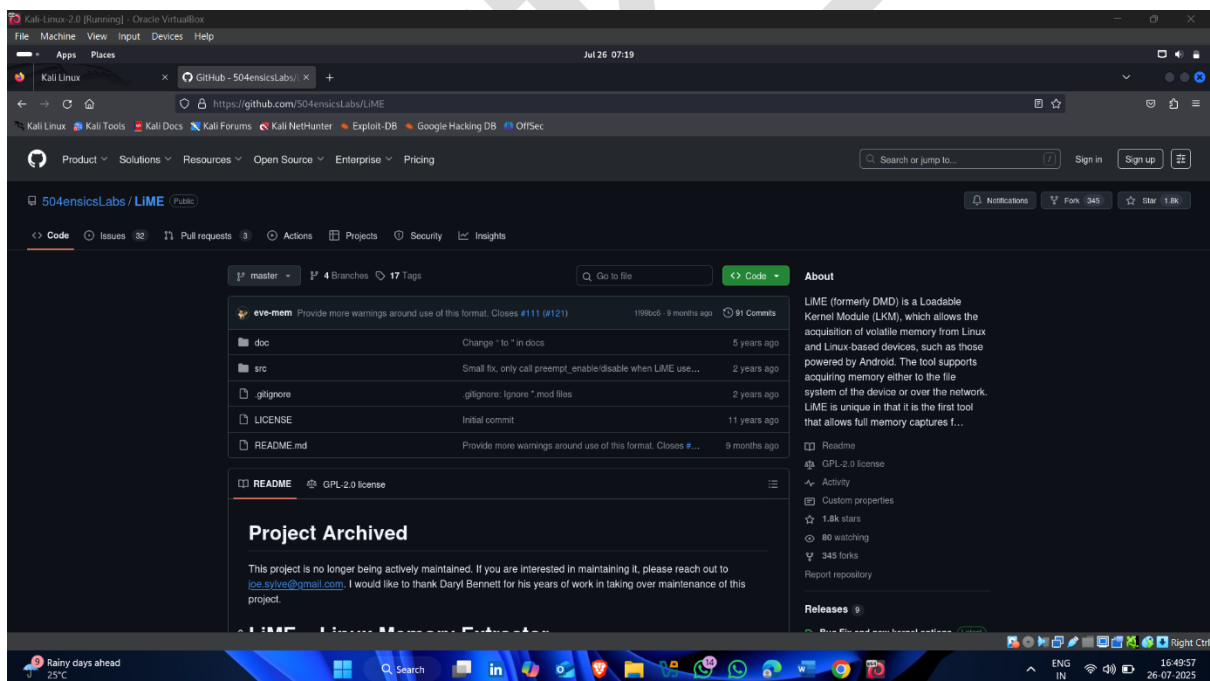
Investigating a **Linux memory dump** (RAM dump) means analyzing a captured snapshot of a system's **volatile memory** (RAM) to extract **evidence**

A) Investigate Forensics Image of Linux Memory Dump Using LiME

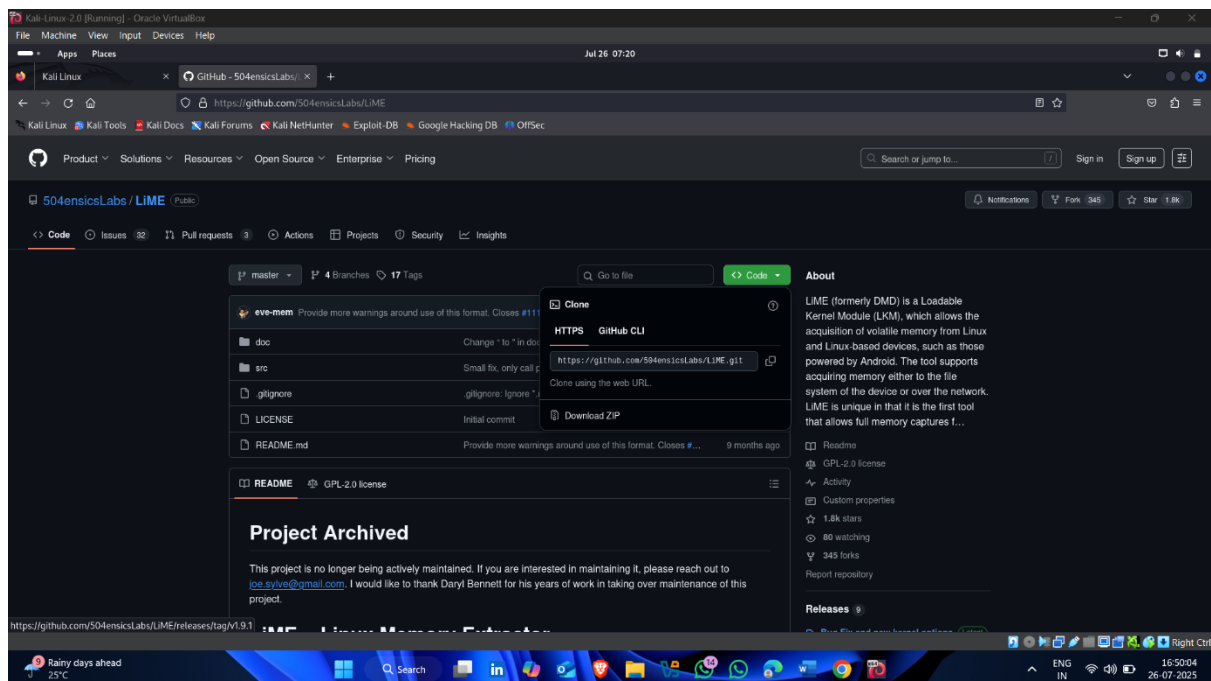
LiME (Linux Memory Extractor) is a **kernel module** used to **acquire volatile memory (RAM)** from a live Linux system. It is one of the most trusted tools for **Linux memory forensics**.

How to use it -:

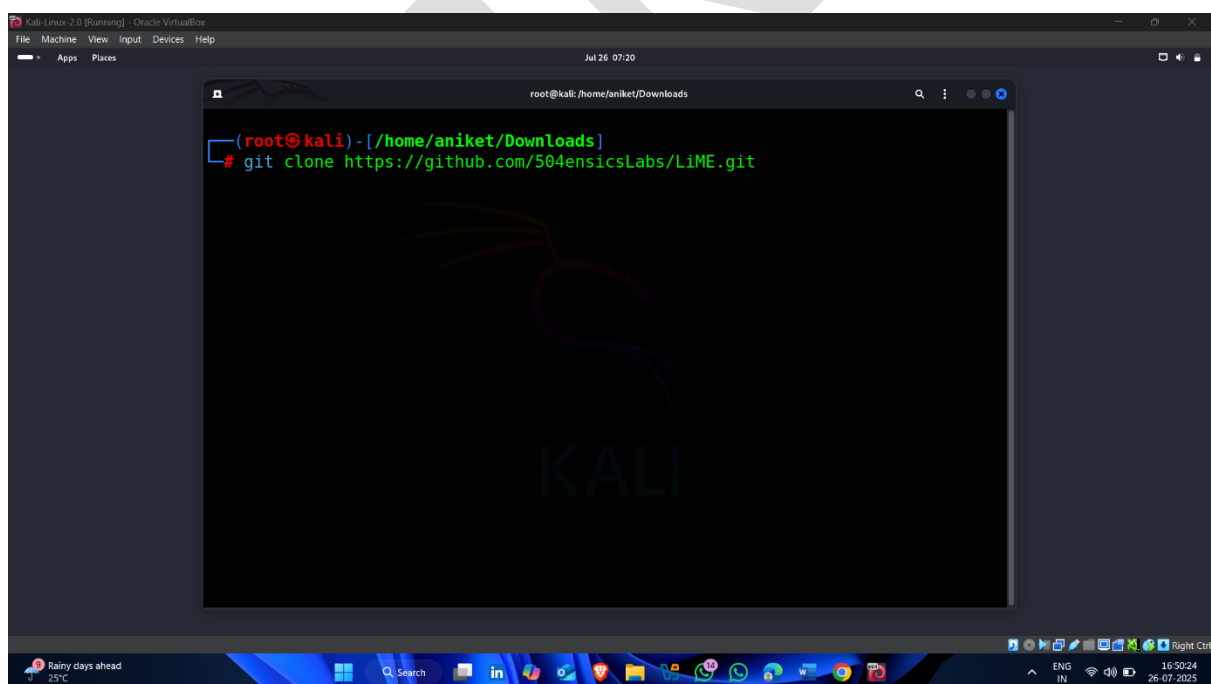
- Click on Green **<Code>** button



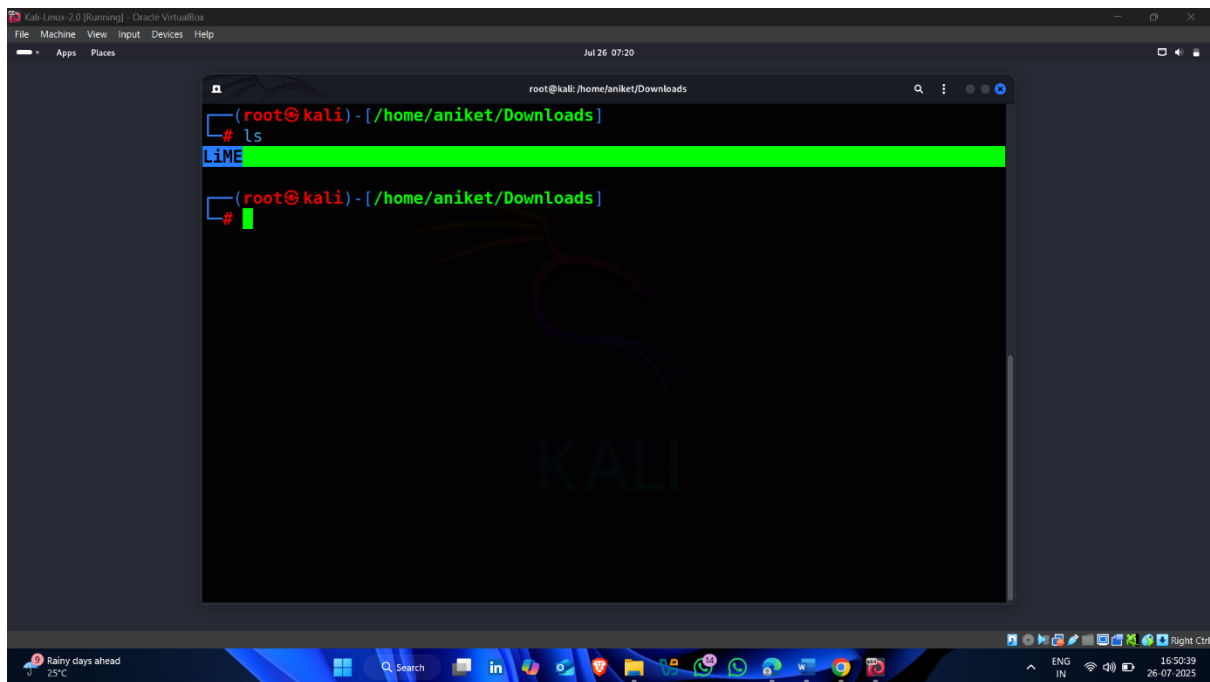
- Copy url



- Paste it on terminal with git clone



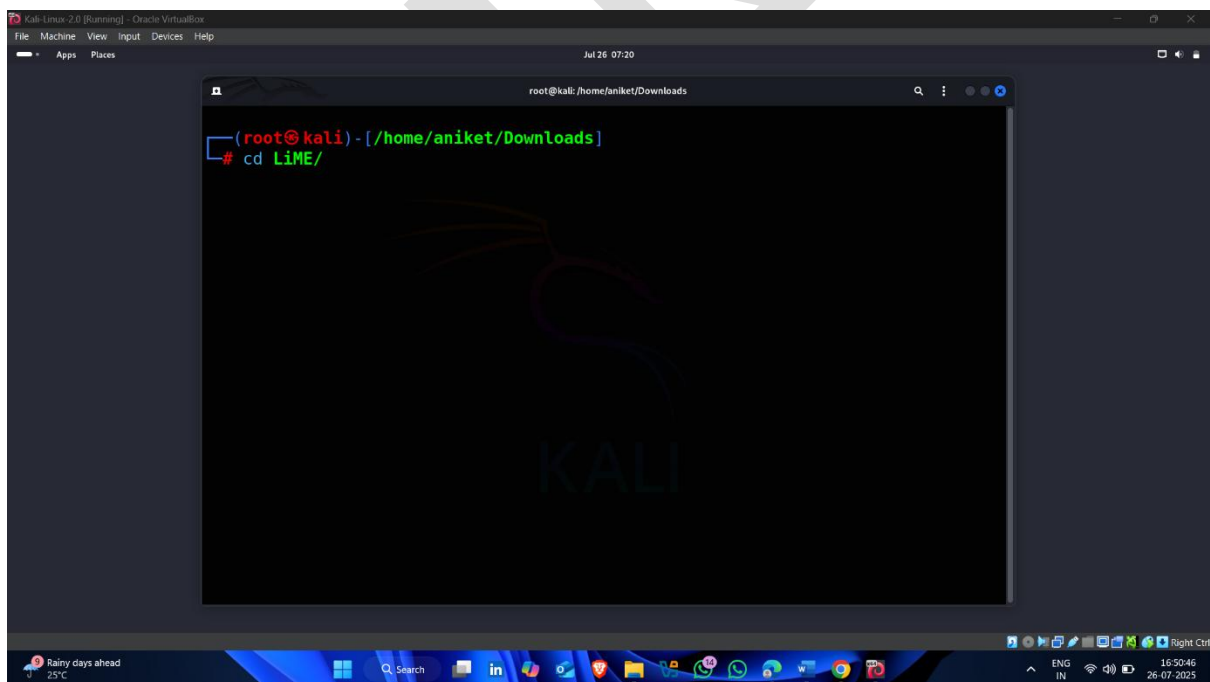
- Download Successfully 📁 ✅



The screenshot shows a Kali Linux terminal window with the title bar "Kali Linux 2.0 [running] - Oracle VM VirtualBox". The terminal prompt is `root@kali: ~/home/aniket/Downloads`. The user has entered the command `ls`, and the output shows a file named `LiME` in green text. The terminal background features a large, faint "KALI" logo. The desktop environment at the bottom includes a taskbar with various application icons and a system tray showing the date and time as 16:50:39 on 26-07-2025.

```
root@kali: ~/home/aniket/Downloads
# ls
LiME
root@kali: ~/home/aniket/Downloads
#
```

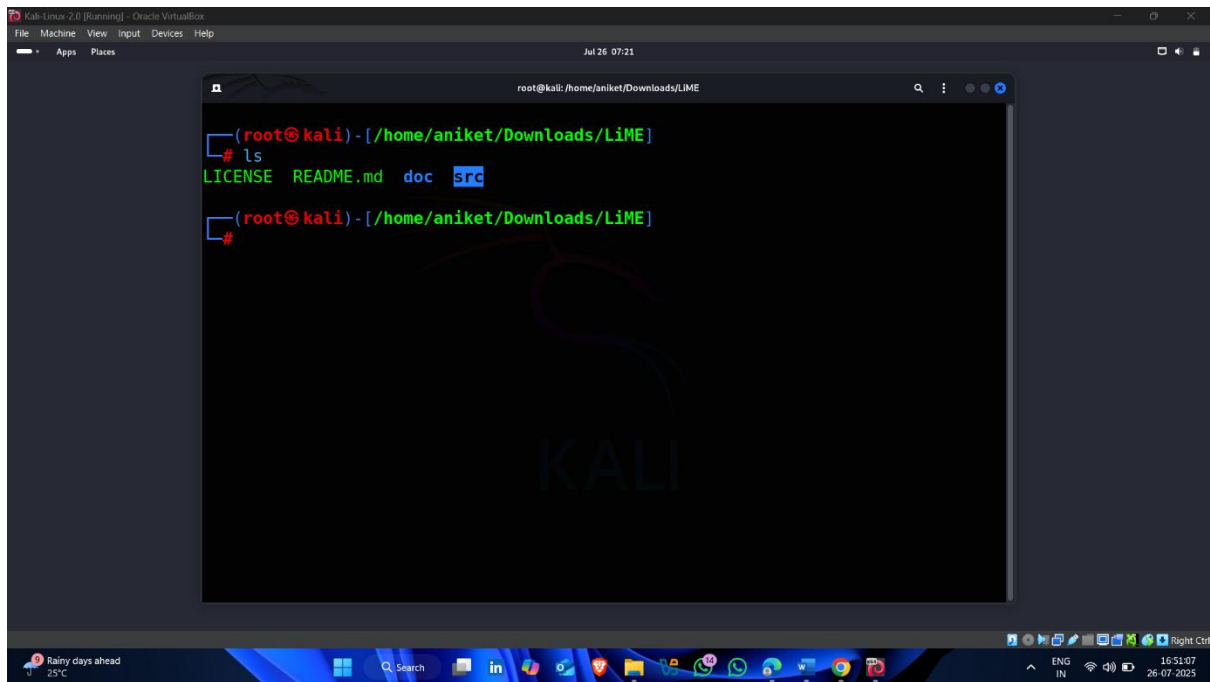
- Now enter `cd` command for enter into LiME Directory



The screenshot shows the same Kali Linux terminal window. The user has entered the command `cd LiME/`, which is highlighted in green. The terminal prompt is now `root@kali: ~/home/aniket/Downloads`. The background and desktop environment are identical to the previous screenshot.

```
root@kali: ~/home/aniket/Downloads
# cd LiME/
root@kali: ~/home/aniket/Downloads
```

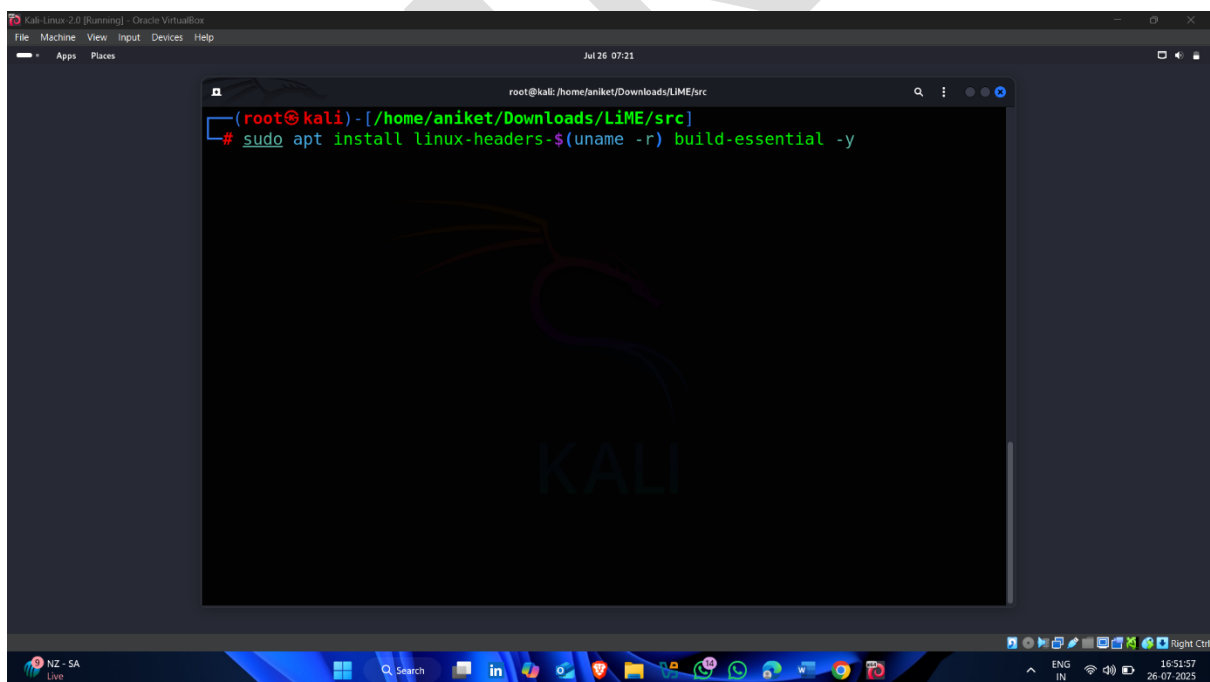
- Now enter **src** Directory Using cd command



The screenshot shows a terminal window titled "root@kali: /home/aniket/Downloads/LiME". The user has entered the command `ls`, and the output is `LICENSE README.md doc src`. The terminal background features a large, faint "KALI" logo. The window is part of an Oracle VM VirtualBox environment, with the host OS being Windows 10. The system clock shows 16:51:07 on 26-07-2025.

```
(root@kali) - [/home/aniket/Downloads/LiME]
# ls
LICENSE README.md doc src
(root@kali) - [/home/aniket/Downloads/LiME]
#
```

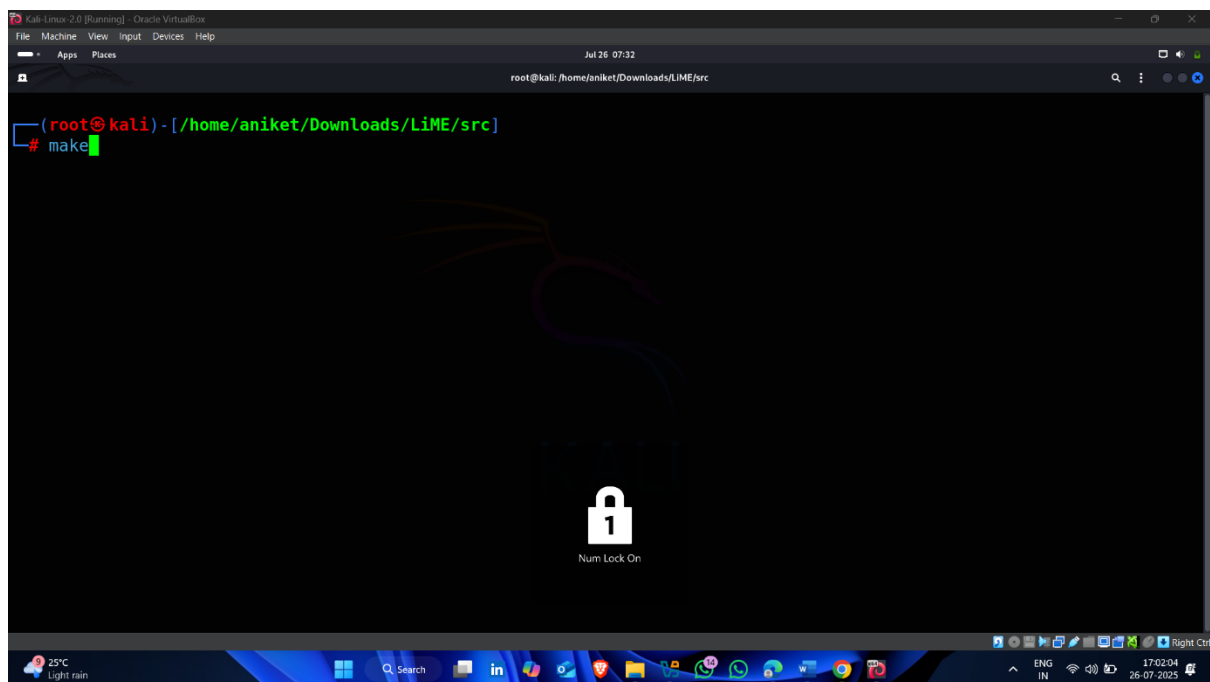
- Now enter following command



The screenshot shows the same terminal window, but the user has navigated to the `src` directory. The prompt now is `(root@kali) - [/home/aniket/Downloads/LiME/src]`. The user has entered the command `sudo apt install linux-headers-$(uname -r) build-essential -y`. The terminal background still features the "KALI" logo. The window title is now `root@kali: /home/aniket/Downloads/LiME/src`. The system clock shows 16:51:57 on 26-07-2025.

```
(root@kali) - [/home/aniket/Downloads/LiME/src]
# sudo apt install linux-headers-$(uname -r) build-essential -y
```

- Now used next command 🖱️

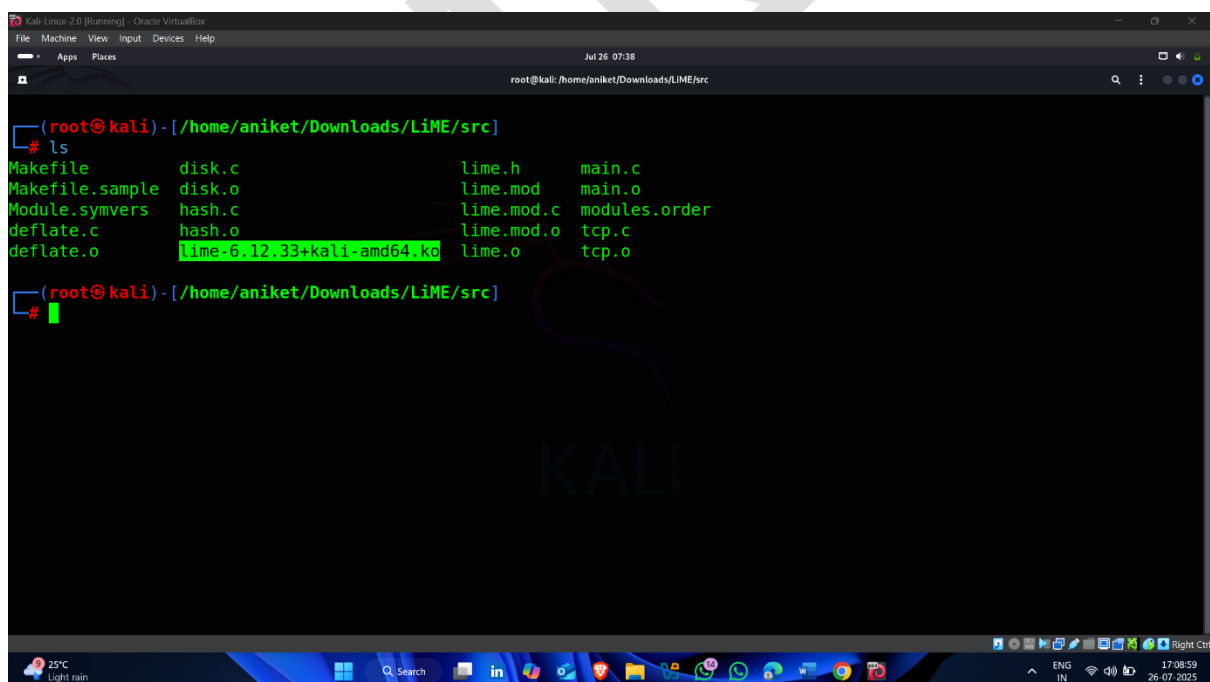


```
Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 26 07:32
root@kali: /home/aniket/Downloads/LiME/src

(root@kali) - [/home/aniket/Downloads/LiME/src]
# make
```

The screenshot shows a Kali Linux terminal window. The prompt is `(root@kali) - [/home/aniket/Downloads/LiME/src]`. The user has entered the command `make`. The terminal output shows a large 'S' logo and a padlock icon with the text 'Num Lock On'.

- Important file ✅



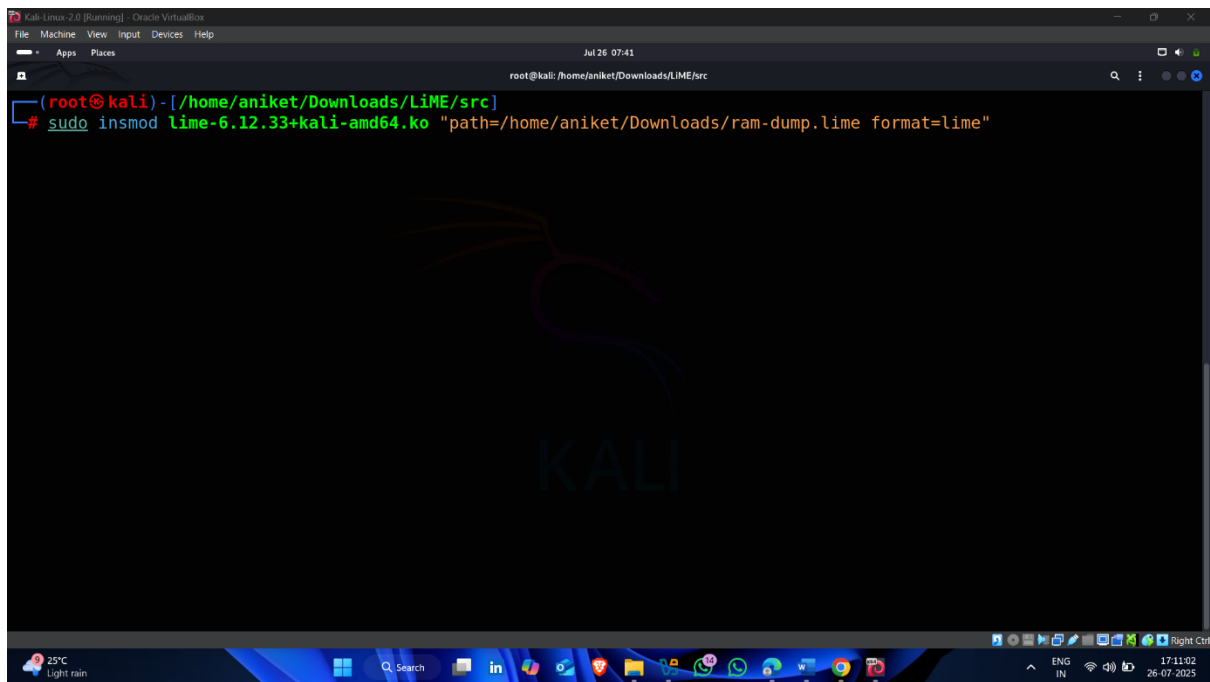
```
Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 26 07:38
root@kali: /home/aniket/Downloads/LiME/src

(root@kali) - [/home/aniket/Downloads/LiME/src]
# ls
Makefile          disk.c            lime.h            main.c
Makefile.sample  disk.o            lime.mod          main.o
Module.symvers    hash.c            lime.mod.c        modules.order
deflate.c         hash.o            lime.mod.o        tcp.c
deflate.o         lime-6.12.33+kali-amd64.ko lime.o             tcp.o

(root@kali) - [/home/aniket/Downloads/LiME/src]
#
```

The screenshot shows a Kali Linux terminal window. The prompt is `(root@kali) - [/home/aniket/Downloads/LiME/src]`. The user has entered the command `ls`. The terminal output shows a list of files and directories. The file `lime-6.12.33+kali-amd64.ko` is highlighted in green.

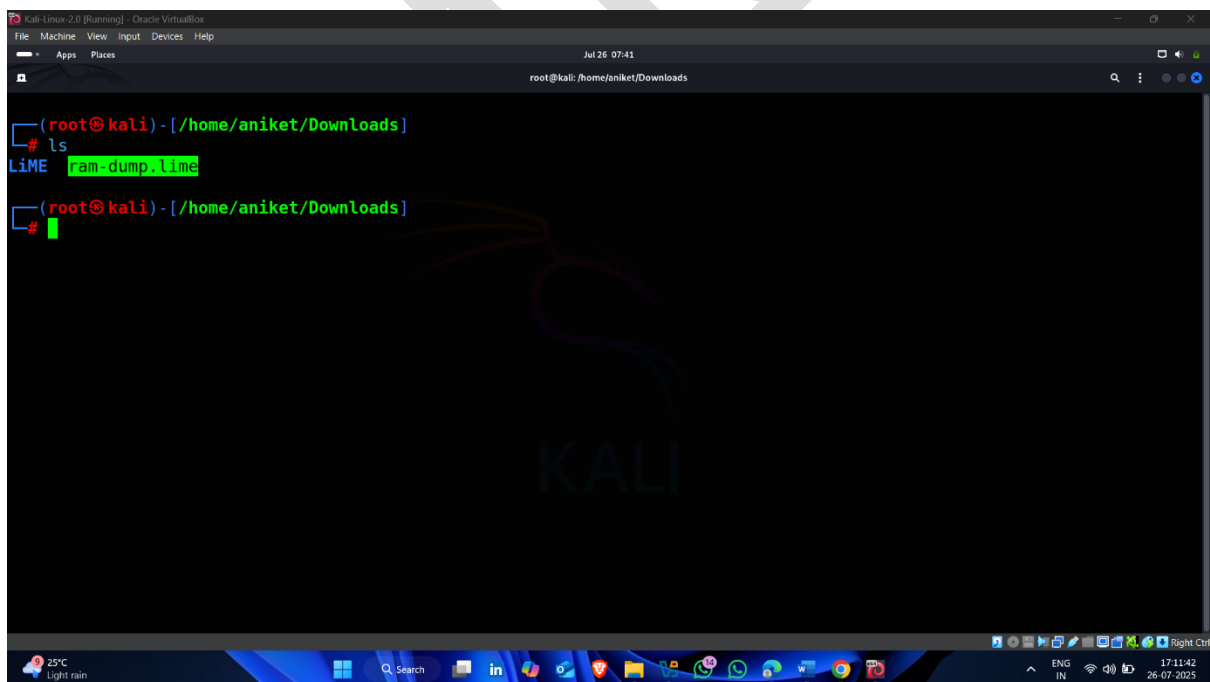
- Now enter following command 🖱️



A screenshot of a Kali Linux terminal window. The window title is "Kali Linux 2.0 [running] - Oracle VM VirtualBox". The terminal shows the user is at the root prompt in the directory `/home/aniket/Downloads/LiME/src`. The command entered is `sudo insmod lime-6.12.33+kali-amd64.ko "path=/home/aniket/Downloads/ram-dump.lime format=lime"`. The terminal background features a large, faint "KALI" logo. The bottom of the window shows a taskbar with various application icons and system status information.

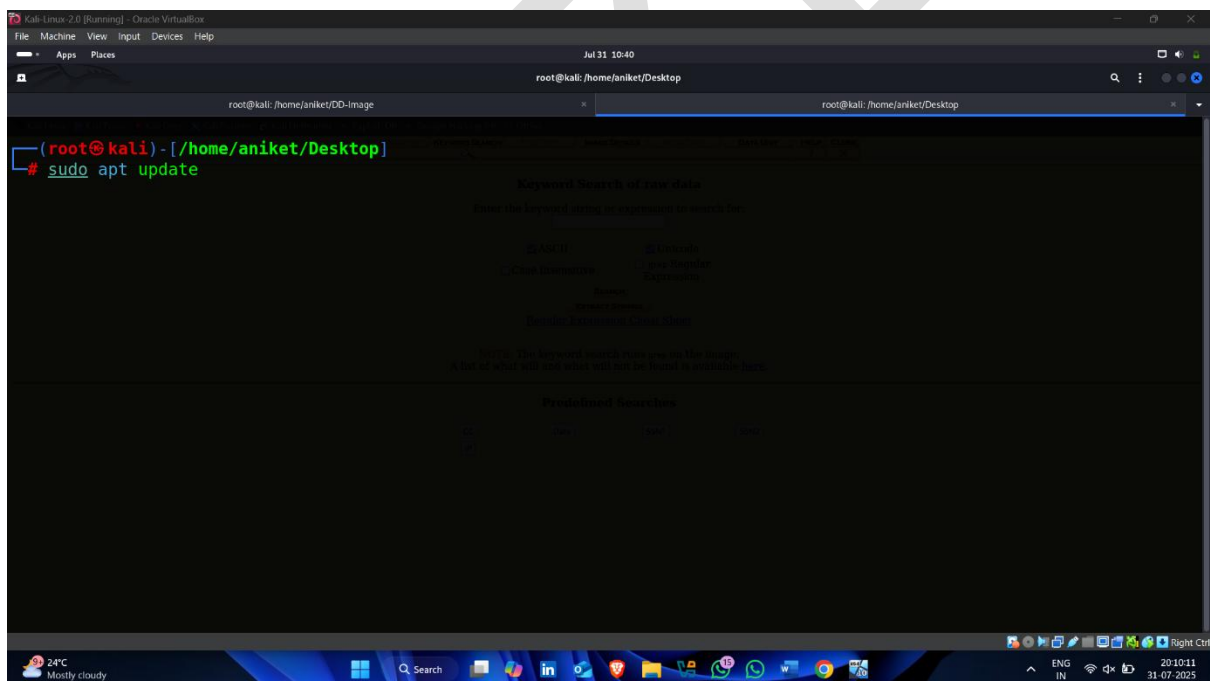
```
(root@kali) - [/home/aniket/Downloads/LiME/src]
# sudo insmod lime-6.12.33+kali-amd64.ko "path=/home/aniket/Downloads/ram-dump.lime format=lime"
```

- Ram-capture file using Lime Tool created successfully 🖱️ ✅



A screenshot of a Kali Linux terminal window. The window title is "Kali Linux 2.0 [running] - Oracle VM VirtualBox". The terminal shows the user is at the root prompt in the directory `/home/aniket/Downloads`. The command `ls` is entered, and the output is `LiME ram-dump.lime`. The terminal background features a large, faint "KALI" logo. The bottom of the window shows a taskbar with various application icons and system status information.

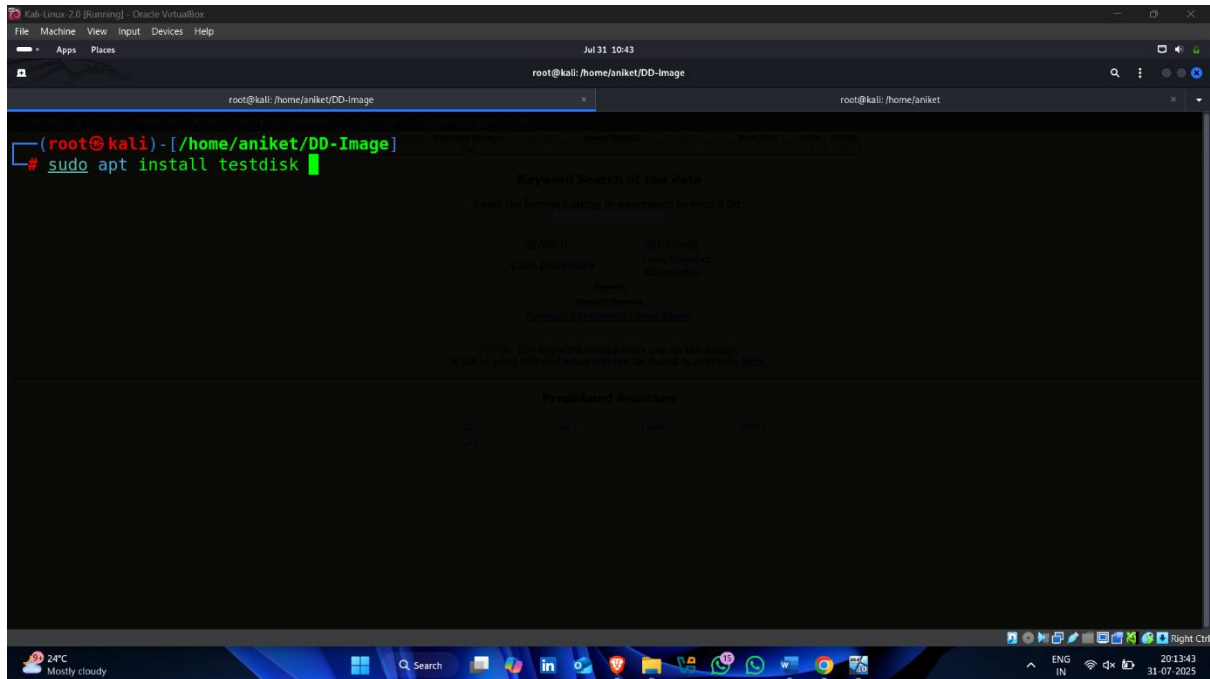
```
(root@kali) - [/home/aniket/Downloads]
# ls
LiME ram-dump.lime
(root@kali) - [/home/aniket/Downloads]
#
```



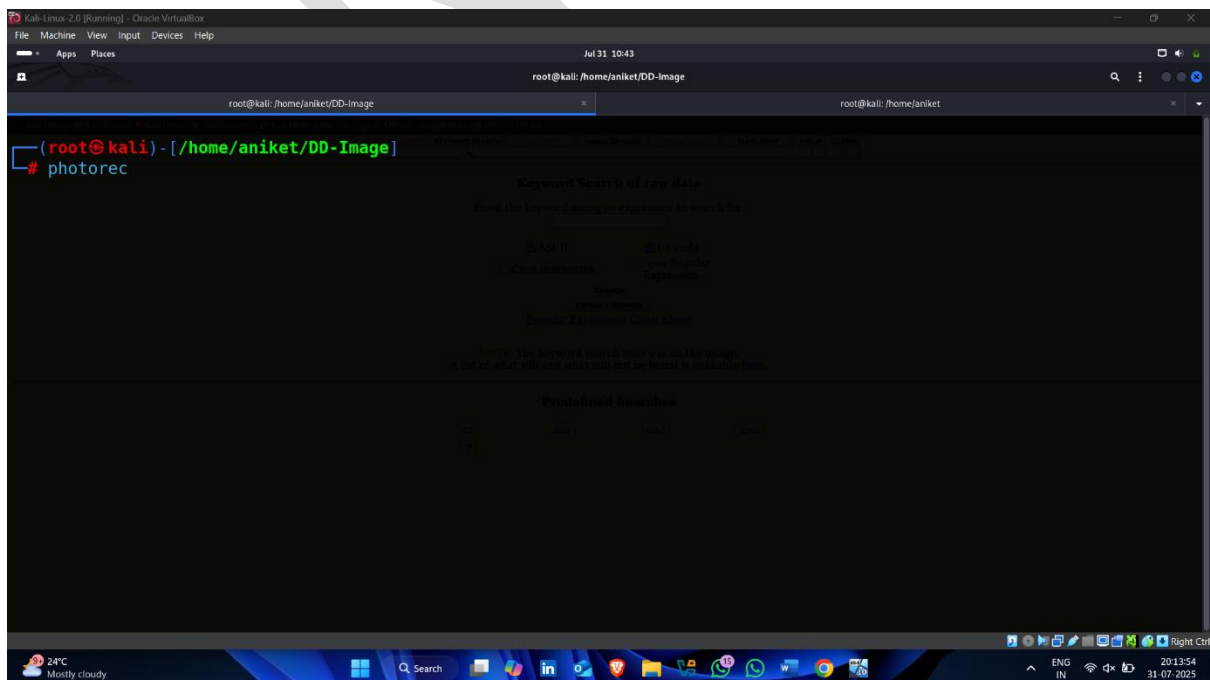
- Then , use next command

Command -: `sudo apt install testdisk`

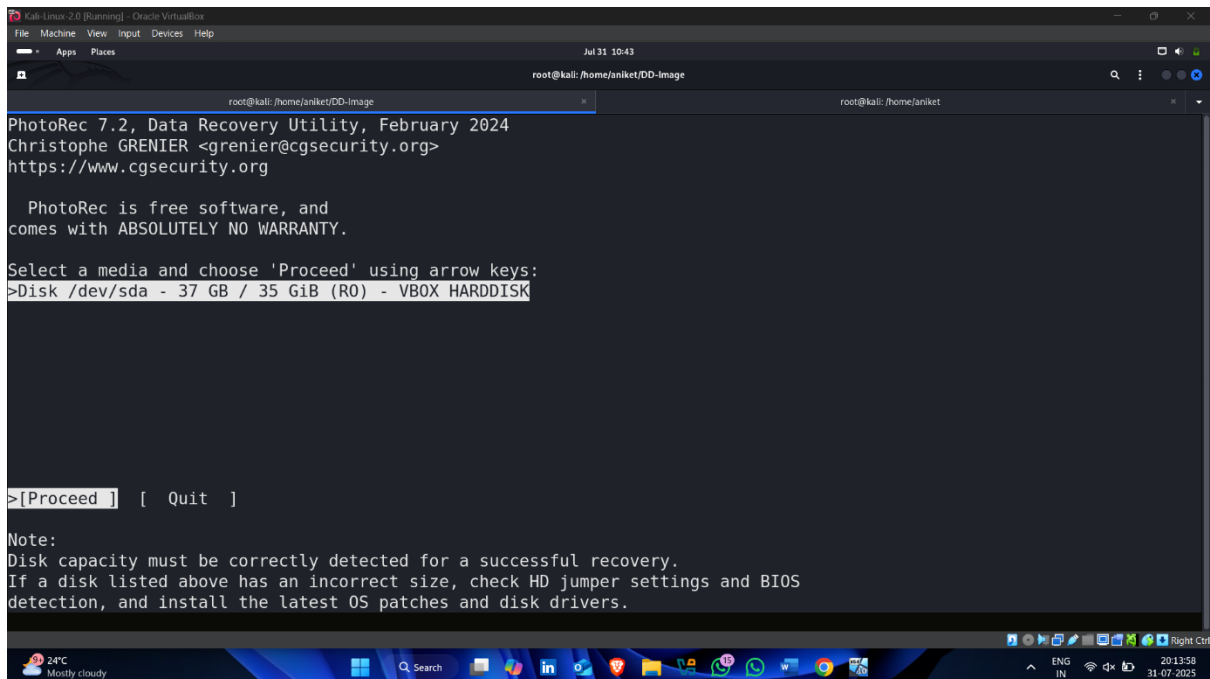
Explanation -: is used on **Debian-based Linux systems** (like **Kali**, **Ubuntu**, **Parrot**, etc.) to **install the TestDisk utility**, which also includes the powerful **PhotoRec** recovery tool.



- Now enter **Photorec** command



- Select **disk** and hit enter for **proceed**



```
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

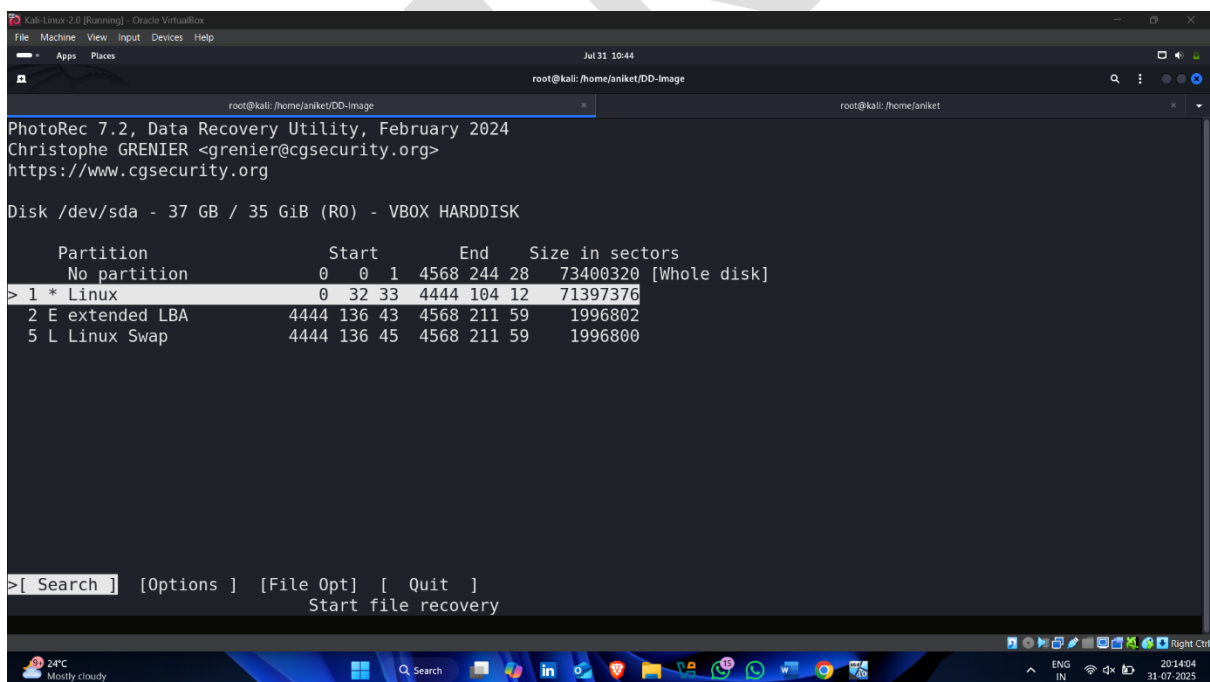
PhotoRec is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk /dev/sda - 37 GB / 35 GiB (R0) - VBOX HARDDISK

>[Proceed] [Quit]
```

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.

- Select Partition



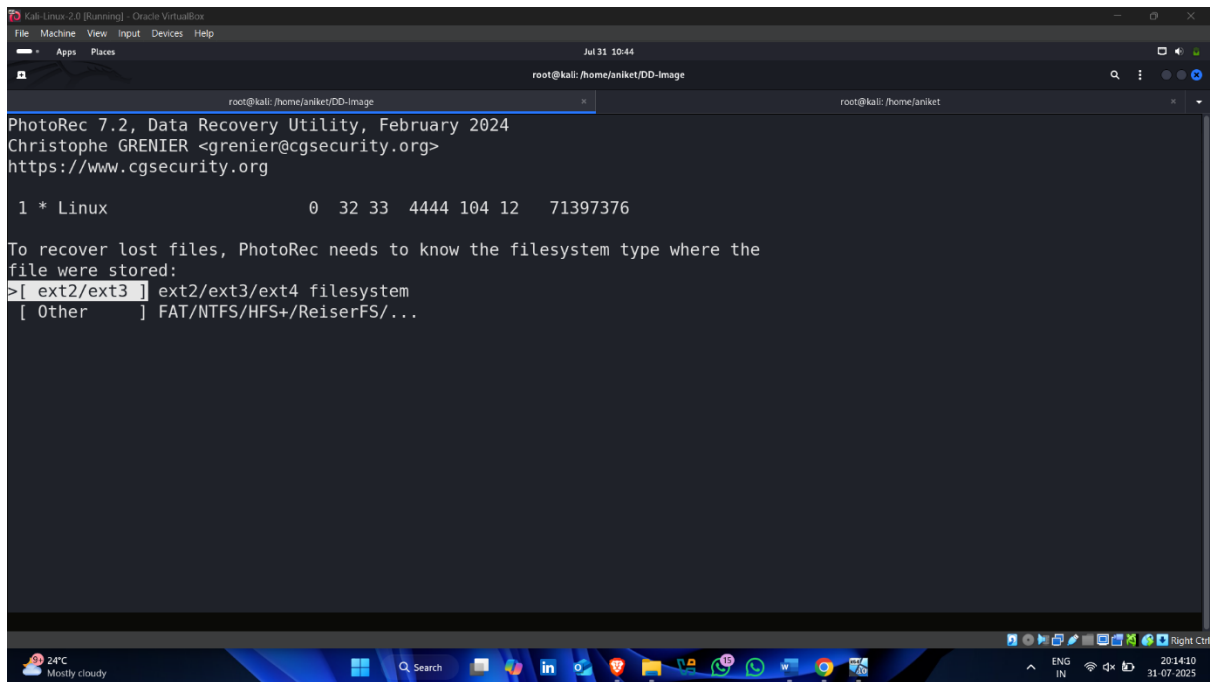
```
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/sda - 37 GB / 35 GiB (R0) - VBOX HARDDISK

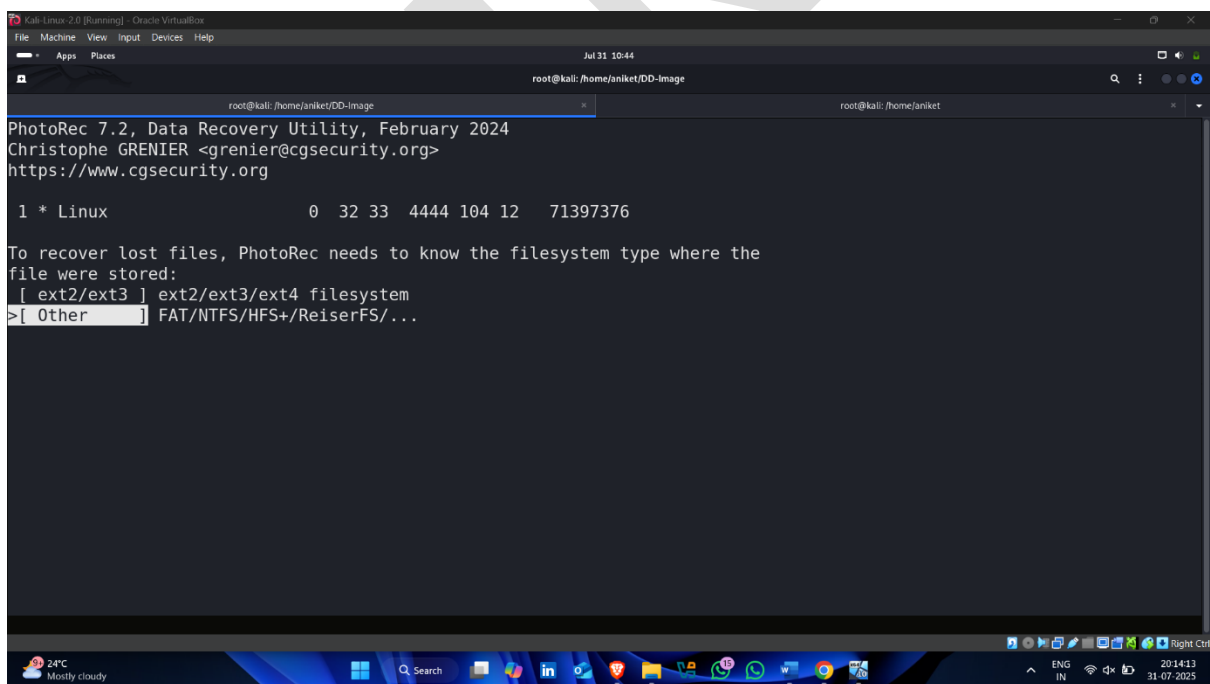
Partition          Start      End      Size in sectors
No partition       0  0  1  4568 244 28  73400320 [Whole disk]
> 1 * Linux         0  32 33  4444 104 12  71397376
  2 E extended LBA  4444 136 43  4568 211 59  1996802
  5 L Linux Swap    4444 136 45  4568 211 59  1996800

>[Search] [Options] [File Opt] [Quit]
Start file recovery
```

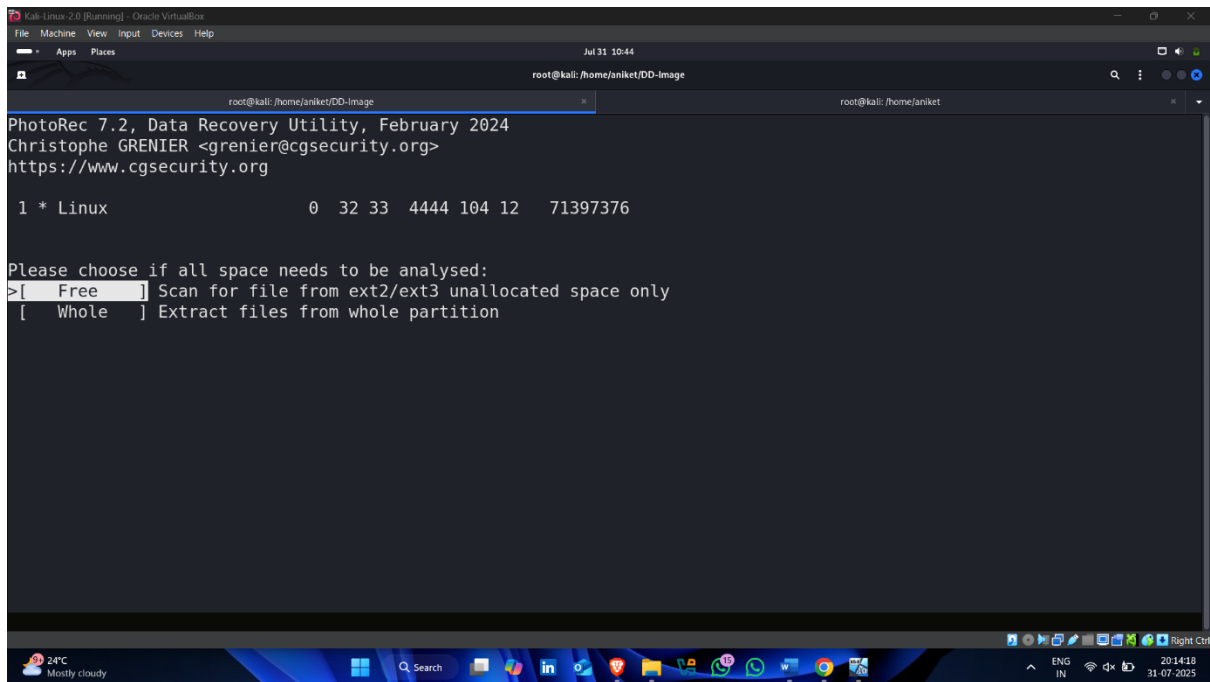
- **Select File type**



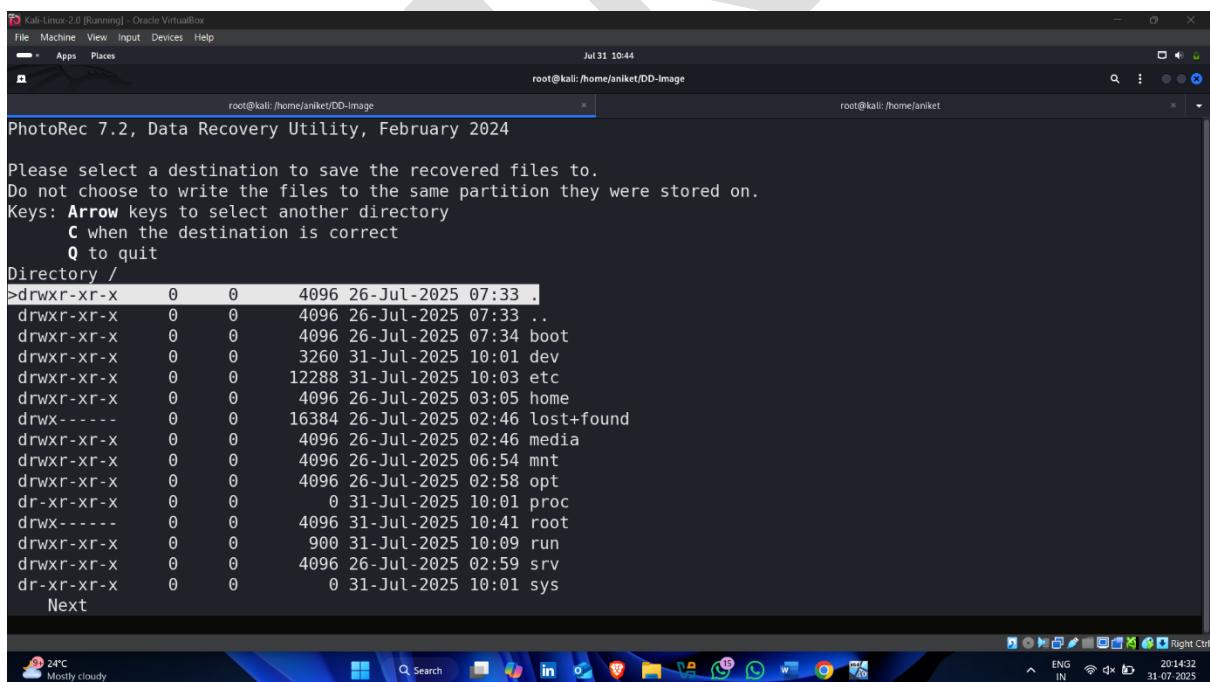
- **Select other**



- Select **First** Option



- Now select dd file location – step by step



```
Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 31 10:44
root@kali: /home/aniket/DD-Image
root@kali: /home/aniket

PhotoRec 7.2, Data Recovery Utility, February 2024

Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home
>drwxr-xr-x  0  0      4096 26-Jul-2025 03:05 .
drwxr-xr-x  0  0      4096 26-Jul-2025 07:33 ..
drwx----- 1000 1000   4096 31-Jul-2025 10:43 aniket
```

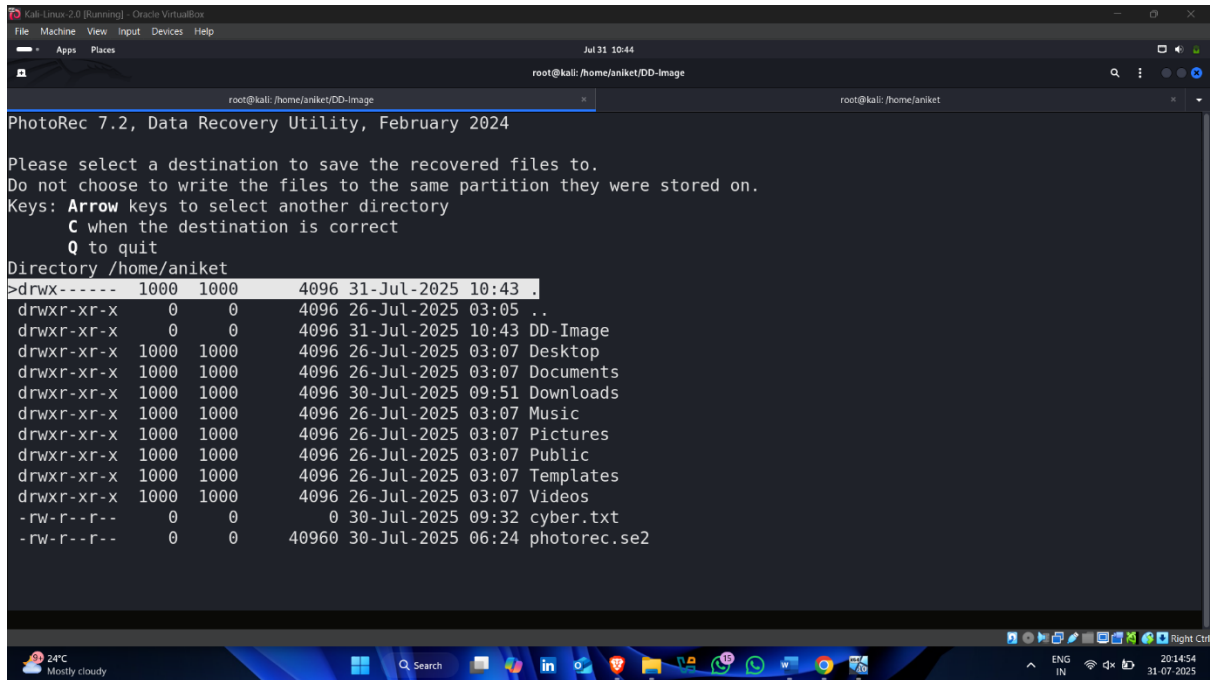
- Select and Hit enter

```
Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 31 10:44
root@kali: /home/aniket/DD-Image
root@kali: /home/aniket

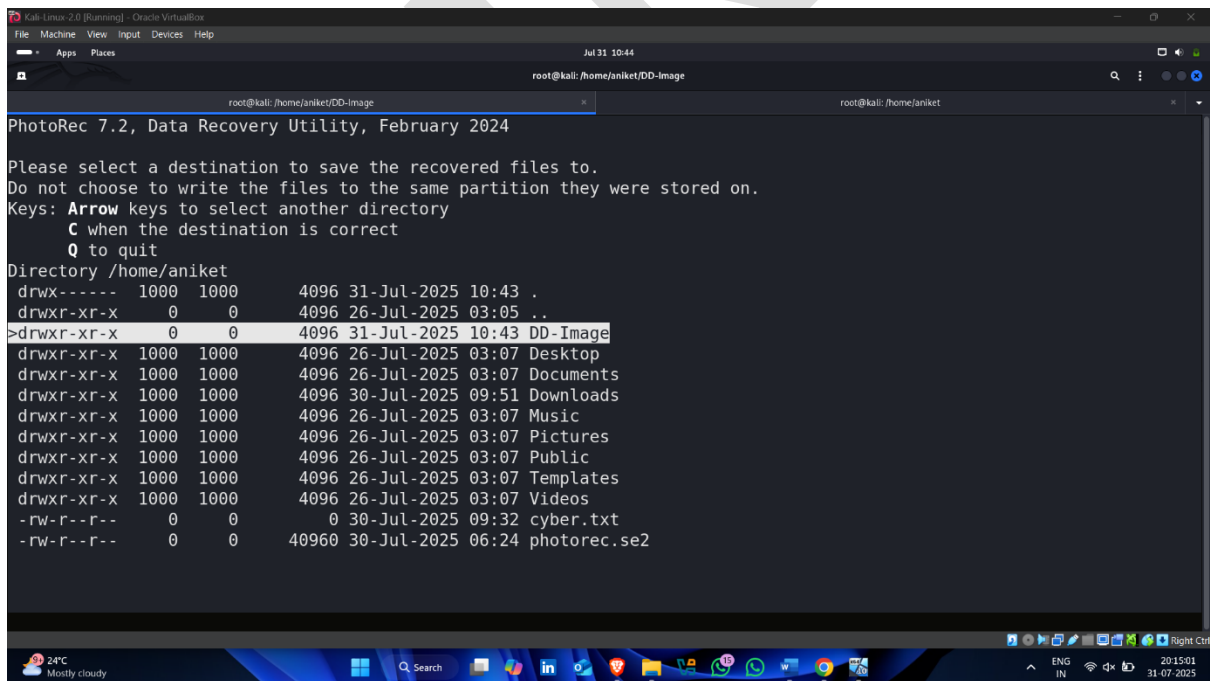
PhotoRec 7.2, Data Recovery Utility, February 2024

Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home
drwxr-xr-x  0  0      4096 26-Jul-2025 03:05 .
drwxr-xr-x  0  0      4096 26-Jul-2025 07:33 ..
>drwx----- 1000 1000   4096 31-Jul-2025 10:43 aniket
```

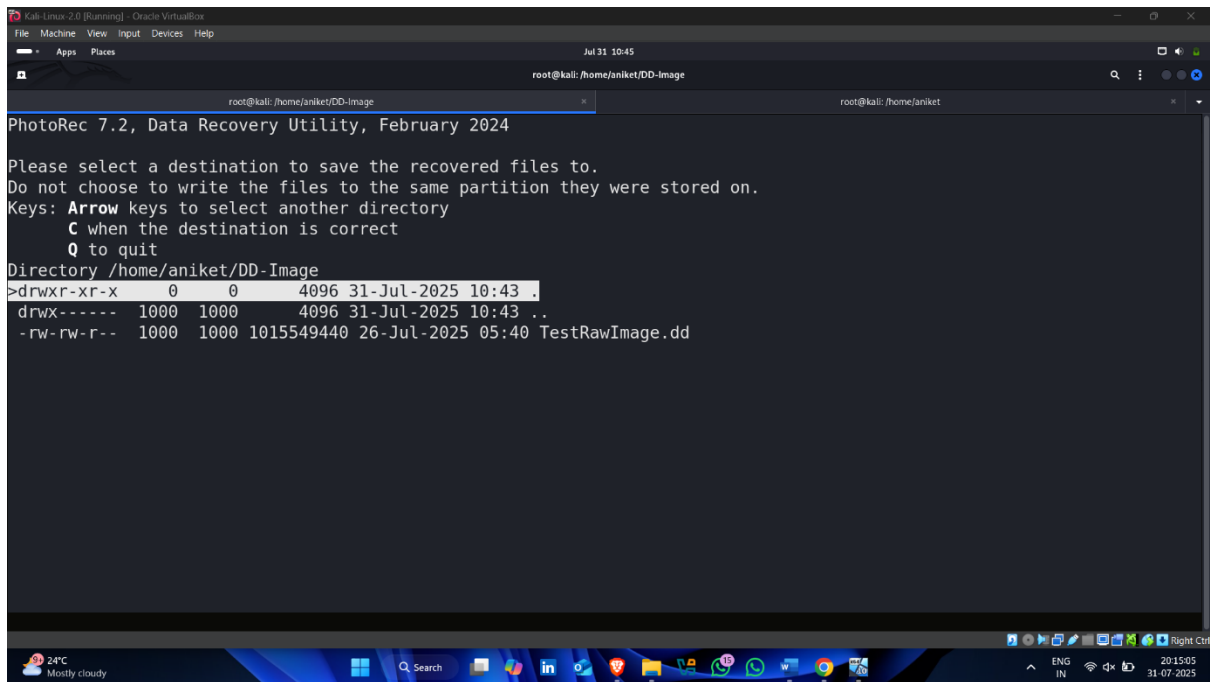
- **Select Download Folder**



- **Hit Enter**



- Now select **dd** extension file

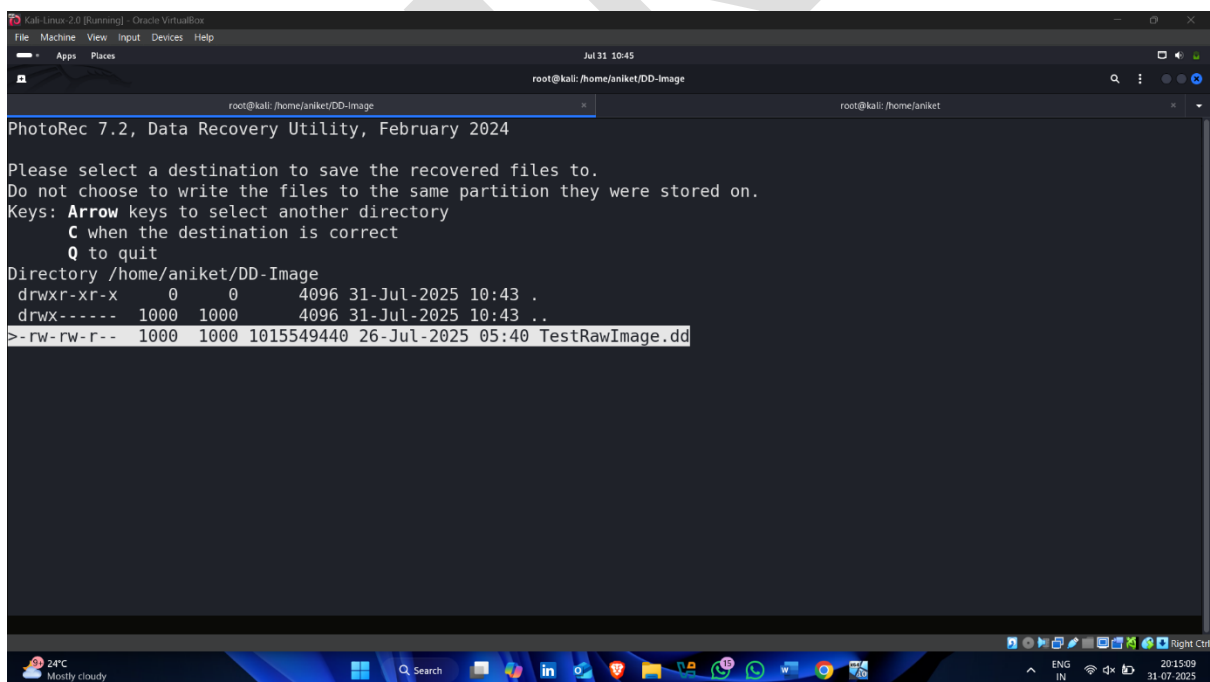


```
Kali Linux 2.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 31 10:45
root@kali: /home/aniket/DD-Image
root@kali: /home/aniket

PhotoRec 7.2, Data Recovery Utility, February 2024

Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home/aniket/DD-Image
>drwxr-xr-x  0  0      4096 31-Jul-2025 10:43 .
drwx----- 1000 1000    4096 31-Jul-2025 10:43 ..
-rw-rw-r-- 1000 1000 1015549440 26-Jul-2025 05:40 TestRawImage.dd
```

- Hit Enter

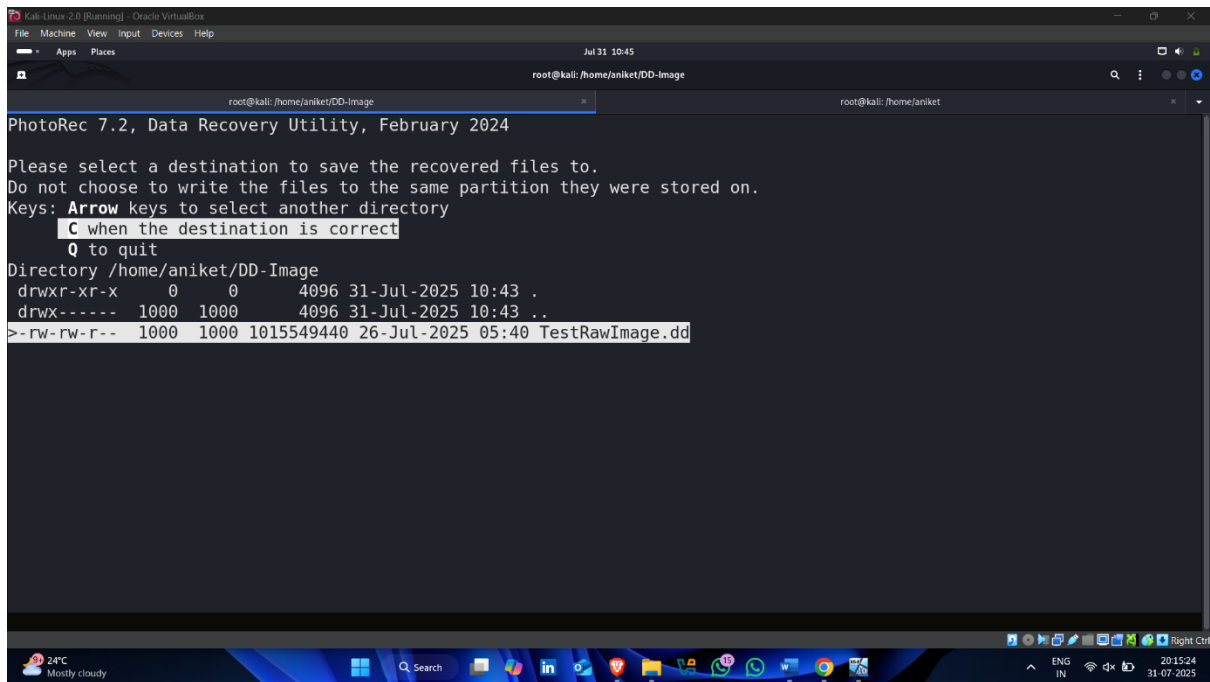


```
Kali Linux 2.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 31 10:45
root@kali: /home/aniket/DD-Image
root@kali: /home/aniket

PhotoRec 7.2, Data Recovery Utility, February 2024

Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home/aniket/DD-Image
drwxr-xr-x  0  0      4096 31-Jul-2025 10:43 .
drwx----- 1000 1000    4096 31-Jul-2025 10:43 ..
>-rw-rw-r-- 1000 1000 1015549440 26-Jul-2025 05:40 TestRawImage.dd
```

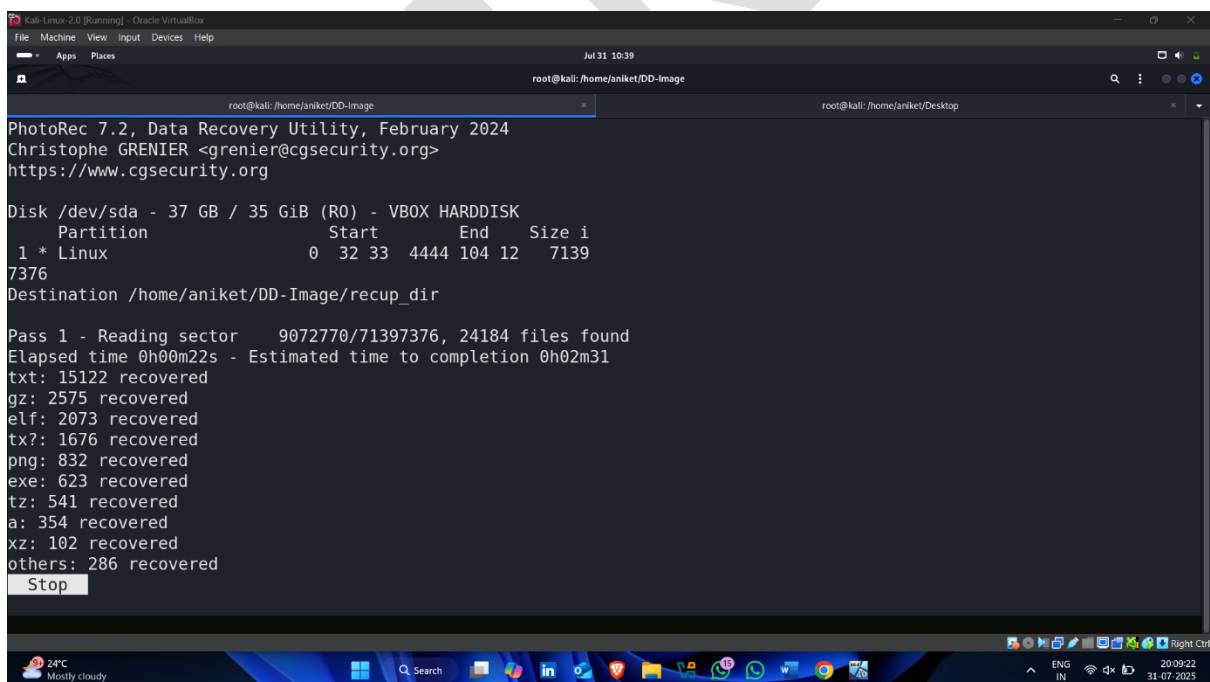
- After select dd image, **press C (Capital C)**



The screenshot shows a terminal window for PhotoRec 7.2. The user is in the directory /home/aniket/DD-Image. The terminal displays the following text:

```
PhotoRec 7.2, Data Recovery Utility, February 2024
Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home/aniket/DD-Image
drwxr-xr-x  0      0      4096 31-Jul-2025 10:43 .
drwx----- 1000 1000      4096 31-Jul-2025 10:43 ..
>-rwx-rw-r-- 1000 1000 1015549440 26-Jul-2025 05:40 TestRawImage.dd
```

- Data recovery started



The screenshot shows the same terminal window as before, but now displaying the recovery progress. The user has pressed 'C' to confirm the destination. The terminal displays the following text:

```
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/sda - 37 GB / 35 GiB (R0) - VBOX HARDDISK
Partition      Start      End      Size i
1 * Linux      0 32 33 4444 104 12 7139
7376
Destination /home/aniket/DD-Image/recup_dir

Pass 1 - Reading sector 9072770/71397376, 24184 files found
Elapsed time 0h00m22s - Estimated time to completion 0h02m31
txt: 15122 recovered
gz: 2575 recovered
elf: 2073 recovered
tx?: 1676 recovered
png: 832 recovered
exe: 623 recovered
tz: 541 recovered
a: 354 recovered
xz: 102 recovered
others: 286 recovered
Stop
```


- Data Recover

```
Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 31 10:40
root@kali: /home/aniket/DD-Image

# ls
DD-Image Desktop Documents Downloads Music Pictures Public Templates Videos cyber.txt photorec.se2

(root@kali) - [/home/aniket]
# cd DD-Image

(root@kali) - [/home/aniket/DD-Image]
# ls
TestRawImage.dd  recup_dir.120  recup_dir.145  recup_dir.17  recup_dir.194  recup_dir.30  recup_dir.55  recup_dir.8
photorec.se2     recup_dir.121  recup_dir.146  recup_dir.170  recup_dir.195  recup_dir.31  recup_dir.56  recup_dir.80
photorec.se2     recup_dir.122  recup_dir.147  recup_dir.171  recup_dir.196  recup_dir.32  recup_dir.57  recup_dir.81
recup_dir.1      recup_dir.123  recup_dir.148  recup_dir.172  recup_dir.197  recup_dir.33  recup_dir.58  recup_dir.82
recup_dir.10     recup_dir.124  recup_dir.149  recup_dir.173  recup_dir.198  recup_dir.34  recup_dir.59  recup_dir.83
recup_dir.100    recup_dir.125  recup_dir.15  recup_dir.174  recup_dir.199  recup_dir.35  recup_dir.6  recup_dir.84
recup_dir.101    recup_dir.126  recup_dir.150  recup_dir.175  recup_dir.2  recup_dir.36  recup_dir.60  recup_dir.85
recup_dir.102    recup_dir.127  recup_dir.151  recup_dir.176  recup_dir.20  recup_dir.37  recup_dir.61  recup_dir.86
recup_dir.103    recup_dir.128  recup_dir.152  recup_dir.177  recup_dir.200  recup_dir.38  recup_dir.62  recup_dir.87
recup_dir.104    recup_dir.129  recup_dir.153  recup_dir.178  recup_dir.201  recup_dir.39  recup_dir.63  recup_dir.88
recup_dir.105    recup_dir.13  recup_dir.154  recup_dir.179  recup_dir.202  recup_dir.4  recup_dir.64  recup_dir.89
recup_dir.106    recup_dir.130  recup_dir.155  recup_dir.18  recup_dir.203  recup_dir.40  recup_dir.65  recup_dir.9
recup_dir.107    recup_dir.131  recup_dir.156  recup_dir.180  recup_dir.204  recup_dir.41  recup_dir.66  recup_dir.90
recup_dir.108    recup_dir.132  recup_dir.157  recup_dir.181  recup_dir.205  recup_dir.42  recup_dir.67  recup_dir.91
recup_dir.109    recup_dir.133  recup_dir.158  recup_dir.182  recup_dir.206  recup_dir.43  recup_dir.68  recup_dir.92
recup_dir.11     recup_dir.134  recup_dir.159  recup_dir.183  recup_dir.207  recup_dir.44  recup_dir.69  recup_dir.93
recup_dir.110    recup_dir.135  recup_dir.16  recup_dir.184  recup_dir.208  recup_dir.45  recup_dir.7  recup_dir.94
recup_dir.111    recup_dir.136  recup_dir.160  recup_dir.185  recup_dir.21  recup_dir.46  recup_dir.70  recup_dir.95
```

```
Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 31 10:42
root@kali: /home/aniket/DD-Image/recup_dir.1

recup_dir.30/ recup_dir.60/ recup_dir.90/ recup_dir.120/ recup_dir.150/ recup_dir.180/
(root@kali) - [/home/aniket/DD-Image]
# cd recup_dir.1

(root@kali) - [/home/aniket/DD-Image/recup_dir.1]
# ls
f0086454.txt f0302848.gz f0343808.elf f0367912.gz f0368584.gz f0425472.elf f0491008.elf f0534480.png
f0086462.txt f0307200.elf f0344064.elf f0367928.gz f0368600.gz f0425984.gz f0491264.elf f0534560.png
f0087082.txt f0307456.elf f0347648.gz f0367944.gz f0368608.gz f0426496.elf f0491520.elf f0534680.png
f0102854.txt f0307712.elf f0348160.elf f0367960.gz f0368616.gz f0427008.elf f0496128.elf f0534688.png
f0103276.txt f0307968.elf f0352768.elf f0367968.gz f0368624.gz f0429056.elf f0496640.elf f0534696.png
f0119606.txt f0308224.elf f0353280.gz f0367984.gz f0368632.gz f0430080.elf f0497152.elf f0534744.png
f0151622.txt f0308480.elf f0354304.elf f0367992.gz f0372736.elf f0432128.gz f0497664.elf f0534808.png
f0152004.txt f0308736.elf f0355328.elf f0368000.gz f0380416.gz f0432896.elf f0498688.elf f0534824.txt
f0168390.txt f0309248.elf f0356352.elf f0368008.gz f0380672.elf f0433152.elf f0499456.elf f0536576.elf
f0169004.txt f0309760.elf f0366856.gz f0368016.gz f0380928.elf f0434176.elf f0499712.elf f0536760.sh
f0169514.txt f0310016.elf f0366944.pl f0368024.gz f0381184.elf f0435712.gz f0507904.elf f0536768.sh
f0169876.txt f0310272.gz f0366976.pl f0368032.gz f0381440.elf f0437248.elf f0511232.elf f0536776.txt
f0171502.txt f0311040.elf f0366984.pl f0368040.gz f0381696.elf f0438272.elf f0511488.elf f0536784.txt
f0176370.txt f0312064.elf f0367008.pl f0368048.gz f0381952.elf f0441856.gz f0512000.elf f0536792.txt
f0188446.txt f0312320.elf f0367072.pl f0368056.gz f0382208.elf f0447488.elf f0515584.elf f0536800.txt
f0205668.txt f0312832.gz f0367080.pl f0368064.gz f0382464.elf f0448512.elf f0516096.elf f0536816.txt
f0207850.txt f0314368.gz f0367096.pl f0368072.gz f0382976.elf f0450304.pm f0516608.elf f0536840.elf
f0221044.txt f0314880.elf f0367104.pl f0368096.gz f0383232.elf f0450560.elf f0517120.elf f0536872.elf
f0221396.txt f0316416.elf f0367144.pl f0368104.qz f0383488.elf f0452608.elf f0518912.elf f0536912.qz
```